

OSCE Coach

Ideal Answers Compendium

Model answers for every active past-question station, grouped by discipline.

Oral surgery

PQ-4

Oral surgery · 11 minutes · Cluster 2

Management of Crown Fracture During Extraction

PQ-6

Oral surgery · 11 minutes · Cluster 3

The station. Mr Parker 50 male, came to your surgery on Friday afternoon, he lives in two hours driving distance from the practice, with pain and swelling on upper molar. On examination 16/26 was badly decayed. You attempted extraction, the crown broke at gum level. Inform the pt about broken crown and discuss the next step of the treatment. pre operative X-ray given showing molar roots are close proximity to sinus, pt taking blood pressure medicine, under control.

An excellent candidate immediately stops the procedure, places gauze over the socket, and calmly informs the patient that the crown of the tooth has broken away during the extraction, which is a recognised complication, particularly with heavily decayed teeth. They explain in plain language that the roots are still present in the jaw and need to be removed, but that this requires a different, more careful approach. The candidate reassures the patient that she is still numb, that they will keep her comfortable, and that they are in control of the situation. They then review the radiograph, noting three divergent, bulbous roots in close proximity to the maxillary sinus floor, and examine the socket for visible root fragments, mobility, and any signs of oroantral communication — specifically asking whether the patient has noticed any air or fluid passing through, and inspecting the socket visually. Given the complex root anatomy, sinus proximity, and the risk of oroantral communication, the candidate explains that the safest next step is either a surgical approach involving a small gum flap and root sectioning if within their competence, or referral to an oral surgeon.

They discuss the realistic risk of an oroantral communication or root slipping into the sinus given the root apices' proximity to the sinus floor, and explain that if this occurs, management depends on the size: small communications can be managed conservatively with a suture and sinus precautions; larger ones require specialist repair root slipping management involves cad well luc approach by the specialist or leaving it there in less than 5 mm and not infected. The candidate addresses the patient's questions about pain (she will remain numb for now; post-operative analgesia such as ibuprofen and paracetamol will be prescribed), cost (private health insurance will likely contribute; they will provide an estimate), and timeframe. They document the complication in the clinical notes, obtain verbal informed consent for the next step, and provide written post-operative and sinus precaution instructions if proceeding, or a referral letter if referring.

appropriate follow up on call after 24 hours instruction and prescription

Management of Post-Surgical Infection Following Extraction

PQ-13

Oral surgery · 11 minutes · Cluster 3

The station. Mrs Joshi is a regular patient at your clinic. You removed her upper left first molar (26) which was badly carious on Monday afternoon and today is Friday afternoon and she booked an emergency appointment, complaining about pain, swelling and bad breath which started 2 days after tooth was removed. After examining her mouth, you are suspecting it's a post-surgical infection. Manage patient's condition, explain the reason and further investigation that needs to be done. MH: Allergic to penicillin.

An excellent candidate would begin by warmly greeting Mrs Joshi and acknowledging her discomfort before taking a focused history. They would establish the timeline: extraction on Monday, symptoms beginning Wednesday, now Friday — approximately 48 hours of progressive pain, swelling, and halitosis. They would ask about the nature and progression of swelling, any fever, difficulty breathing or swallowing, trismus, and what analgesia she has tried. They would take a thorough medical history and confirming allergy to penicillin. They would ask about her temperature, noting she recorded 37.8°C this morning.

On examination, the candidate would perform a thorough extra-oral assessment, noting the indurated, warm, erythematous swelling over the left mandibular body and submandibular region, measuring mouth opening at approximately 30mm indicating mild trismus, and critically checking for floor-of-mouth elevation, dysphagia, or any signs of airway compromise — all absent here. Intra-orally, they would inspect the socket for purulent discharge and assess surrounding tissues.

The candidate would diagnose a post-operative odontogenic infection and explain this clearly to Mrs Joshi in plain language, reassuring her while being honest about the need for prompt treatment. They would explain that diabetes impairs immune response and wound healing, increasing infection risk, and advise her to monitor blood glucose closely and contact her GP.

open disclosure accept that even though its a rare complication i should have informed you about it.

explain your management today(if temprature is more than 38 will require antibiotics) identify and remove the cause (if we are not able to identify the cause refer to hospital) xray for record and cinform if no rootpiece left behind

Given penicillin allergy, the candidate would prescribe cephalexin if allergy was mild or consider clindamycin 300mg TDS if allergy was severe. They would arrange review in 24–48 hours (if the swelling starts to improve continue treatment and if it doesnt improve refer to specialist suspecting oeteomyelitis) and provide clear safety-net advice: if she develops difficulty breathing or swallowing, floor-of-mouth swelling, rapidly worsening trismus, or high fever, she must present immediately to a hospital emergency department. Further investigations would include OPG to assess the socket and exclude retained root fragments.

Management of a Patient on Prednisone and Alendronate Requiring Extraction PQ-14

Oral surgery · 11 minutes · Cluster 1

The station. *Mrs Xiao is a 55-year-old patient attended your surgery today, complaining of badly decayed and she has been getting pain on 37 tooth for last 10 days. The tooth cannot be saved and needs extraction. Medically she is fit and well, she is taking Actonel tablets of 150mg once a month for osteoporosis condition for the last 4 years. She is also taking Prednisolone 10mg for her rheumatoid arthritis for the last 7 years. Please take consent for extraction and address patients concerns.*

An excellent candidate would begin by warmly greeting Mrs William, confirming her identity and the reason for her visit, then take a focused medication history. They would clarify that she is taking Actonel (risedronate) 150 mg monthly for osteoporosis for four years, and Prednisolone 10 mg daily for rheumatoid arthritis for seven years. They would note the concurrent use of a bisphosphonate and a corticosteroid as two independent risk modifiers for the planned extraction.

The candidate would then explain the risk of medication-related osteonecrosis of the jaw (MRONJ) in plain language: that bisphosphonates affect how bone heals after surgery, risk is increased by concurrent corticosteroid use and the duration of therapy exceeding four years. They would reassure Mrs Xiao that thats why it is preferred to do it with the specialist.

Next, the candidate would identify adrenal suppression risk. Seven years of Prednisolone 10 mg daily is sufficient to suppress the hypothalamic-pituitary-adrenal axis. Extraction under local anaesthesia is a moderate surgical stressor, and the candidate would discuss the need to liaise with her prescribing doctor regarding perioperative steroid supplementation.

For the management plan, the candidate would explain that the specialist would extract as indicated, with appropriate precautions: atraumatic surgical technique, primary wound closure where possible, chlorhexidine rinses, and consideration of prophylactic antibiotics.

Finally, they would provide smoking cessation advice, prescription for pain, and review

Management of Post-Operative Haematoma Following Dental Treatment PQ-16

Oral surgery · 11 minutes · Cluster 2

The station. Heather James is regular attendee at your clinic. booked an emergency appointment today complaining of swelling, mild discomfort and limited mouth opening following the IANB that was given a week before because of composite filling. You did examination today and you noticed patient can open up to 20mm, with swelling. You gave the diagnosis of trismus, hematoma and myospasm. Explain the diagnosis and how will you manage pts concern.

An excellent candidate would begin by warmly greeting Mrs Hayes and acknowledging her concern about the alarming appearance of the swelling and bruising trismus. They would confirm any changes in medical history, confirm the length and the difficulty of the procedure, and the timeline of swelling onset — noting it began within two hours and progressed overnight with bruising now tracking toward the neck.

They would specifically ask about medications, identifying aspirin 100mg daily and perindopril, and confirm she was correctly advised to continue aspirin. They would screen for red flag symptoms: any difficulty breathing, difficulty swallowing, worsening pain, fever, or rapidly increasing swelling — all of which are absent here. On examination, they would note extraoral ecchymosis and swelling, assess mouth opening nearly 20mm. make the patient understand why you gave a block anesthesia and not infiltration. The candidate would then explain the diagnosis clearly: the extraction caused minor bleeding from small blood vessels into the surrounding soft tissues, and because aspirin reduces platelet function, the blood spread more widely than usual, producing the bruising now visible on the face and neck. They would reassure Heather James that while the bruising looks frightening, it is not dangerous in her situation — it is self-limiting and will resolve over one to two weeks, changing colour from purple to yellow as it does.

Management advice for Hematoma would include warm compresses now to aid reabsorption 20 minutes / hour, continuing regular paracetamol for mild discomfort,

management for trismus maintaining a soft diet, and keeping the area clean.

The candidate would safety-net clearly: Mrs James should return immediately or go to an emergency department if she develops any difficulty breathing or swallowing, rapidly increasing swelling, fever above 38°C, or signs of infection such as pus or worsening pain.

They would offer a review appointment in five to seven days to assess Hematoma and Trismus if improving continue with the treatment if hematoma is not improving refer to oral surgeon (may need MRI to assess where the blood is, risk of infection and scarring) if trismus is not improving refer to oral medicine specialist to assess for TMD

torres islander nurse PQ-19

Oral surgery · 11 minutes · Cluster 3

The station. Torre islander nurse comes today complaining of pain from 16. You did the examination and 16 has irreversible pulpitis and insisted on pulling it out. Now you have decided to extract the tooth. Xray shows that 16 roots are close to the sinus and there is an infection around 1 root. Patient is worried about OAC, Hepatitis B and HIV infection as she heard of a breach in some other dental clinic. manage patients concerns.

An excellent candidate would begin by introducing themselves, demonstrating cultural respect from the outset. They would take a focused medical history, absence of allergies, and regular GP contact at the Aboriginal Medical Service. explains the xray The candidate would then assess the severity of the dental infection by

asking about fever, swelling progression, systemic symptoms, recognising that while the infection is localised, it requires urgent management. They would explain clearly that tooth 16 infected and needs to remove the source of infection

Demonstrates knowledge of current Australian guidelines: Standard protocol set by DBA to prevent HIV and Hep B infection spread usually they spread by sharps injury during procedures. Sharps management (to prevent injuries) reusable instruments are transferred safely in lidded tray sterilized as per their category. non critical and critical. Critical instrument protocol includes washed / ultrasonic bath/ pouched/ BCI/ chemical indicators inside every pouch in every cycle / goes into patients notes / opened at point of use to ensure safety.

understanding of Australian system treatment free for Torres Islanders in community clinics and in private clinic treatment is chargeable but they are provided vouchers and specialist treatment available at a discount.

The candidate would address the proximity of roots to the sinus, explaining the risk of oro-antral communication and how they would manage this if it occurred.

management prescribe pain medicines (Antibiotic needed if it is apical periodontitis and you are not treating today) short term tooth removal/RCT/temporisation (pulpectomy) long term replacement options

use health promotion to consider why everyone in family extracts teeth (maybe they don't have access to treatment)

Consent for Extraction PQ-25

Oral surgery · 11 minutes · Cluster 3

The station. Mr Patrick is a regular patient at your clinic. Tooth 16 is badly broken down and unrestorable — it needs extraction. X-ray provided showing tooth 16 (upper right first molar) grossly carious and unrestorable, with the palatal roots in close proximity to the floor of the maxillary sinus. Explain the procedure, describe the complications (common and serious), give post-operative instructions, and confirm the patient's understanding before proceeding.

The candidate begins by confirming the patient's identity and the tooth to be extracted, clearly stating 'tooth 16, the upper right first molar.' They acknowledge Mr Patrick's anxiety and reassure him that his comfort is the priority throughout. The candidate explains the procedure in plain language: local anaesthetic will be given to numb the area, the tooth and surrounding gum will be gently loosened, and the tooth will be removed using dental instruments. They note the procedure typically takes 15–30 minutes. The candidate explains why extraction is recommended — the tooth is too badly broken to restore canal blocked as per xray — and briefly mentions that root canal treatment was considered but is not viable given the extent of the damage.

Common risks are discussed clearly: some pain and swelling for a few days, possible bruising, minor bleeding, difficulty opening the mouth, and a small risk of damage to adjacent teeth. Serious risks specific to this tooth are addressed: dry socket (alveolar osteitis), oro-antral communication given the proximity of the upper molar roots to the maxillary sinus, possible root fracture and slip into sinus requiring surgical retrieval, and that nerve injury is rare for this tooth.

OAC management less than 2 mm absorbent gel / suture / heals more than 3 mm referral and management by specialist slip try to remove with suction if not out can leave it if not infected or preferred to manage by oral surgeon Cadwell Luc approach

The candidate addresses Mr Patrick's smoking history directly, advising him to avoid smoking for at least 48 as it significantly increases the risk of dry socket. Post-operative instructions include biting on gauze for 30 minutes, avoiding rinsing for 24 hours, taking paracetamol and/or ibuprofen as directed, eating soft foods, and returning if bleeding does not settle or pain worsens after day three. The candidate confirms he can drive home as local anaesthetic alone does not impair driving, and advises he should be able to return to work the following day in most cases. The candidate checks Mr Patrick's understanding, invites questions, and confirms

his voluntary consent before documenting in the clinical notes.

also discusses briefly replacement options for the tooth and explains the need for it. (offer Broucher)

uses his concern of healing in health promotion to explain how smoking impacts healing and invites to quit and offers quitline in an motivational way.

Non-Healing Extraction Socket PQ-32

Oral surgery · 11 minutes · Cluster 1

The station. Mrs Crocket 55 year old female patient had extraction done 4 days ago by another dentist across the road and pain started soon afterwards. Pt complains of bad smell and bad taste. Give possible causes of pain, what investigations will you do and manage patients concern.

An excellent candidate would begin by introducing themselves and establishing rapport with Mrs Crocket

acknowledging her discomfort and the fact that she has been in pain for several days. They would take a focused history covering the timeline of the extraction, the duration and difficulty of the procedure, the onset and character of the pain/ fever, any bad taste or smell, and what analgesia she has already tried. They would specifically ask about her medical history,

Social history Smoking history and Alcohol

On examination, the candidate would inspect the socket at site 45, noting the exposed greyish bone, absence of a blood clot, minimal granulation tissue, halitosis, and tenderness on gentle probing, while confirming the absence of swelling, lymphadenopathy, purulent discharge, noting partial paraesthesia on the lip.

The provisional diagnosis of post operative infection would be clearly stated, and active smoking are significant complicating factors that impair healing. The candidate would explain the condition to Mrs Crocket in plain language — that the protective blood clot has been lost from the socket, leaving the bone exposed and causing the severe pain and odour.

Discuss Risk factors and how each has an impact Smoking / Alcohol(increases bleeding) / debris/ cleaning with tooth pick/

Management Examination extraoral (swelling / temperature / lymphnodes/ jaw joints) and intraoral (socket/photos/ Xray) Nerve test 2 point discrimination/ temperature /touch/ pressure)

management would include gentle irrigation of the socket with warm saline, and upgrading analgesia to ibuprofen 400mg with paracetamol 500–1000mg alternating four-hourly if not contraindicated.

Review 48 hours Review for nerve injury 2 weeks / 1 month / 3 months (if no improvement referral to oral and Maxillofacial surgeon)

The candidate would advise Mrs Chrocket to see her GP to rule out deficiency/ diabetes provide brief smoking cessation advice and offer referral to Quitline (13 7848), and arrange review in 48–72 hours. Safety-netting would include clear instructions to return immediately if she develops increasing swelling, fever, trismus, or difficulty swallowing.

Offer a medical certificate for missed work.

Management of Apical Root Fragment Retained During Extraction PQ-49

Oral surgery · 11 minutes · Cluster 3

The station. 50+ year-old male. Came to the surgery on a Friday afternoon. Lives 2 hours driving distance from the practice. Scenario: Patient presented with pain and swelling on upper right molar side which was badly decayed. Was advised for extraction and agreed. During extraction, palatal root fractured and small apical 1/3 is left behind. Discovered that tooth had a vertical crack after extraction. On radiograph, root was in close proximity to maxillary sinus. Patient is taking blood pressure medication. Currently under control. Clinical Image: Periapical X-ray of the upper molar region showing retained root tip/fragment with a highlighted circle indicating its location close to the maxillary...

An excellent candidate would begin by pausing the procedure, ensuring the patient is comfortable, and calmly disclosing what has occurred. They would explain in plain language that during the extraction a small piece of the root tip — approximately 3 to 4 millimetres — has broken away and remains in the bone near the upper right back tooth area. They would acknowledge the patient's likely concern and normalise the situation by explaining this is a recognised complication of difficult extractions, particularly when teeth have curved or fragile roots or underlying cracks, as was found here. The candidate would then explain the clinical findings: the fragment is small, it is sitting close to the floor of the maxillary sinus, there is no sign of active infection, and the socket does not appear to be communicating with the sinus. They would present two management options clearly. First, attempting immediate surgical retrieval, which carries risks including excessive bone removal, potential perforation of the maxillary sinus, prolonged procedure time, and increased post-operative discomfort. Second, a conservative monitoring approach, which is supported by evidence for small, asymptomatic apical fragments in the absence of infection or sinus communication — such fragments often become encapsulated or resorb without causing problems. Given the fragment size, absence of infection, proximity to the sinus, and the patient's well-controlled blood pressure, the candidate would recommend conservative management with close follow-up. They would involve the patient in this decision, answer her questions honestly, and obtain verbal consent for the monitoring plan. Clear post-operative instructions would be provided: avoid nose-blowing, watch for increasing pain, swelling, discharge, or a bad taste, and return immediately if any of these occur. A review appointment would be arranged in one to two weeks with a follow-up periapical radiograph at three to six months. The incident would be documented fully in the clinical notes including the fragment size, location, radiographic findings, discussion held, and the agreed management plan.

Oroantral Communication — Disclosure After Upper Molar Extraction PQ-55

Oral surgery · 11 minutes · Cluster 3

The station. Mrs Elida Jones, regular patient. Scenario: You performed an extraction of tooth 26 (upper left first molar). During the extraction, the tooth fractured. You successfully removed approximately 2/3 of the tooth but 1/3 of the root (~1–2 mm) remains in the socket. On testing sinus patency (Valsalva / nose-blow test), air is bubbling from the socket — confirming oroantral communication (OAC). Task: Explain to the patient what has happened during the extraction. Inform her about: (1) the retained root fragment, (2) the oroantral communication, and (3) immediate and ongoing management. Address her concerns.

An excellent candidate opens by gently orienting Mrs Jones before delivering the news, signalling that there is something important to discuss. They disclose both complications — the retained root fragment and the oroantral communication — clearly and without delay, using plain language throughout. They explain that the tooth fractured during removal, which is a recognised risk with heavily restored upper molar teeth, and that while most of the tooth was successfully removed, a small root tip of approximately one to two millimetres remains. They explain that attempting to retrieve this fragment now carries a greater risk of pushing it into the sinus than leaving it under specialist review, and that a specialist will assess it within the next day or two. They

then explain the oroantral communication in lay terms — that a small opening has formed between the mouth and the sinus cavity above, confirmed by the bubbling seen when she breathed out through her nose. When Mrs Jones asks 'Did you do something wrong?', the candidate responds honestly: the tooth fractured, this is a known complication particularly with heavily restored upper molars, and while they are genuinely sorry it happened, it does not represent a failure in technique. The candidate asks about penicillin allergy before prescribing amoxicillin 500 mg three times daily for five days, and adds a nasal decongestant spray. They ask about prior sinus problems, elicit the history of resolved chronic sinusitis, and explain why this makes careful follow-up especially important. They give specific OAC precautions — no nose blowing for two weeks, sneezing with the mouth open, no straws, no smoking. They arrange urgent referral to an oral and maxillofacial surgeon within 24 to 48 hours. When Mrs Jones raises concern about living alone, they provide written instructions, a practice after-hours number, and clear criteria for seeking emergency care.

Post-Extraction Instructions for Patient on anticoagulant Apixaban (Eliquis) PQ-61

Oral surgery · 11 minutes · Cluster 3

The station. *Mr Amaro, 70, regular patient. You have just extracted a mobile tooth 23 (upper left canine). Haemostasis gel has been applied and a suture placed. He takes apixaban (Eliquis) to prevent stroke; you did not stop the medication prior to extraction, having consulted with his GP beforehand. He had a previous extraction 3 years ago, before he was started on Eliquis. He smokes 25 cigarettes a day. Task: give him post-extraction instructions and explain the possible complications that may happen.*

This station has two explicit parts — post-extraction instructions, and the complications that may happen — and both need to be covered clearly. The two risk multipliers running through the whole answer are his apixaban and his smoking, and Mr Amaro is genuinely anxious about bleeding specifically, so calm, concrete answers matter as much as the content itself.

OPENING "Mr Amaro, I've removed your tooth and I'm going to go through the instructions you'll need to take care of the wound — are you sitting comfortably? I'll also give you everything I explain in writing, so you don't need to remember it all."

WHY THE MEDICATION WASN'T STOPPED When he asks why his Eliquis wasn't stopped, or suggests pausing it himself: "We deliberately kept you on it, on your GP's advice — for someone taking it to prevent stroke, the risk of stopping it, even briefly, is higher than the bleeding risk from this extraction, which we can manage well with the gel and stitch we've placed. So please keep taking it exactly as normal — don't skip or delay any doses."

EXPLAINING THE GEL AND SUTURE "Apixaban keeps your blood flowing smoothly, which is exactly what you want for preventing a stroke — but it means the blood in the socket doesn't clot as easily as usual. Normally, blood fills the gap left by the tooth and forms a jelly-like clot that protects the bone underneath while it heals. In your case that might not happen as reliably on its own, so we've placed a gel that acts like a scaffold to help the clot form and hold — over time it becomes part of your healing bone. We've also put in a dissolvable stitch to hold things together; it'll come out on its own, no need to have it removed."

ADDRESSING HIS ANXIETY ABOUT BLEEDING When he asks "will I bleed to death?": "That's a really understandable worry, but serious bleeding like that is rare, and there are clear, effective steps if bleeding doesn't settle — I'll walk you through exactly what those are so you know what to expect." If he asks what would actually happen: "First we'd clean the area and identify exactly where it's coming from, then we might inject some local anaesthetic with a medication that helps constrict the blood vessels, and if needed we can cauterise the spot to seal it. Severe bleeding needing something like a blood transfusion is very uncommon and not what we're expecting here."

IMMEDIATE PLAN AND HOME CARE "I'd like you to wait in the waiting room for a bit so I can check the bleeding has fully settled before you head off, and I'll give you written instructions for everything." If oozing happens later, firm pressure with gauze for 20 to 30 minutes, or a slightly damp black or green tea bag as a good alternative — the tannins help it clot. Contact the practice if there's persistent ooze, anything that concerns him, or bleeding that stops and then restarts, especially heavily. A cold pack wrapped in a towel — never directly on the skin — on the cheek for the first 24 hours, about 20 minutes on and 20 off, helps limit swelling and bruising, which may be more noticeable than usual because of the apixaban.

PAIN RELIEF Paracetamol first-line; explicitly rule out ibuprofen: "I'd avoid that — anti-inflammatories add to your bleeding risk on top of the Eliquis." If paracetamol alone isn't controlling it, that's worth flagging rather than just taking more — persistent pain despite paracetamol can be an early sign of something like dry socket.

FIRST 24 HOURS VS AFTER No alcohol, no smoking, no vigorous rinsing or spitting, no hot food or drinks, and avoid strenuous exercise for the first 24 hours. From the next day, gentle warm salt-water rinses are fine.

WHY THIS TIME IS DIFFERENT When he compares to his simple extraction three years ago: "That one went smoothly because you weren't on a blood thinner then, and you're a little older now too — age and ongoing smoking both mean healing isn't quite as fast or predictable as it was back then. Nothing's gone wrong, it's just why I want to be thorough this time."

SMOKING Raised specifically, not as a lecture: "I do want to mention your smoking, because it's genuinely relevant — it's part of why this tooth needed to come out in the first place, since gum disease made it lose its footing, and canines are usually the last teeth people lose. It also raises your risk of a painful complication called dry socket after this extraction, and slows healing generally. If you can avoid smoking for at least 48 to 72 hours, ideally longer, that makes a real difference. If you'd ever like support, Quitline is a good free option, and there's nicotine replacement too — and keeping up with an electric toothbrush and regular hygiene visits will help protect your other teeth long-term."

COMPLICATIONS — STATE THESE EXPLICITLY "A few things worth knowing about: some bleeding today is normal, but bleeding that keeps pooling or restarts heavily needs attention. Because you smoke, you're at higher risk of dry socket — that shows up as a strong ache a few days from now, sometimes with a bad taste or smell, and we treat it with a soothing dressing if it happens. There's a small risk of infection, which looks like worsening pain, swelling or fever rather than things settling down. Bruising and swelling of the cheek is common, especially with the Eliquis, and isn't dangerous by itself — the cold pack helps with that. You might also notice some numbness around the area for a little while, which is usually temporary, but let us know if it doesn't settle. And healing overall may just take a bit longer than last time, given your age and smoking."

IF HE ASKS ABOUT WARFARIN "Apixaban was chosen partly because it avoids some of warfarin's downsides — warfarin interacts with a lot of medications and foods, and needs regular blood tests to make sure the dose is right, which apixaban doesn't require."

SAFETY NET A specific contact point — the practice during hours, or the emergency department for severe uncontrolled bleeding — plus the written instructions and the plan to recheck him before he leaves today.

COMMON WAYS TO FAIL THIS STATION Agreeing to or not correcting his idea of pausing the apixaban. Dismissing his bleeding anxiety with vague reassurance instead of a specific, calm explanation. Not explicitly addressing smoking, or lecturing rather than explaining the specific risk. Failing to explicitly list complications as their own section. Missing dry socket specifically, or not linking it to smoking. Recommending or failing to rule out ibuprofen/NSAIDs. Giving only a vague safety net instead of specific bleeding criteria and a contact point.

Post-Operative Instructions for Patient on Liquid Diet

PQ-42

Oral surgery · 11 minutes · Cluster 3

The station. David Chen, 28-year-old office administrator. Scenario: You have just surgically removed both lower wisdom teeth (38, 48) — raised mucoperiosteal flaps, bone removal with handpiece, sectioning of teeth, and 3-0 Vicryl sutures placed bilaterally. No complications during surgery. Local anaesthetic given (articaine with 1:100,000 adrenaline); patient is still numb. Medical/Social History: No medical conditions, no regular medications. Non-smoker. Lives alone in Parramatta, works full-time in an office, has taken 2 days off work. First time having oral surgery, mildly anxious. Clinical Image: No intraoral clinical photo — text-only slide on dark background. Task: Give post-extraction...

An excellent candidate would begin by confirming what was done and checking the patient's understanding, acknowledging that both lower wisdom teeth were surgically removed today and that the sites are still numb from the local anaesthetic. They would explain that because raised flaps, bone removal, and sutures were involved, the healing sites need protection, which is why a liquid diet is recommended for the first 24 to 48 hours, progressing to soft foods over the following five to seven days. The candidate would give specific examples of suitable foods — cool or lukewarm soups, smoothies, protein shakes, milk, and yoghurt drinks — while advising the patient to avoid hot, cold, or spicy foods that could irritate the sites. They would clearly advise against using straws, explaining that the suction pressure can dislodge the blood clot and lead to dry socket, a painful complication. For general post-operative care, the candidate would advise the patient to avoid rinsing for the first 24 hours to protect the clot, then begin gentle warm salt water rinses from the following day. They would recommend applying ice packs to the cheeks for the first 24 hours to manage swelling. Regarding analgesia, the candidate would confirm that paracetamol 1g every six hours is available at home and would recommend adding ibuprofen 400mg every eight hours with food, taken in alternation, explaining this combination is more effective than either alone. They would advise the patient that two days off work is appropriate given the bilateral surgical nature of the procedure. Safety-net advice would include warning signs requiring urgent review: excessive or uncontrolled bleeding, severe worsening pain after day two or three suggesting dry socket, fever, swelling that is increasing rather than decreasing, or any difficulty swallowing or breathing. The candidate would provide the practice's emergency contact number and confirm the patient has a way to get home safely given they are still numb.

Periodontics

Orthodontic Patient with periodontal problem

PQ-1

Periodontics · 11 minutes · Cluster 1

The station. Jordan 17-year-old boy came to your practice for consultation for crowded teeth. He has seen a dentist 12 months ago. You did CPITN scoring today (Community Periodontal Index of Treatment Needs) which is 444/424 and you have taken bitewing x-ray showing bone loss around molars. Intraoral picture showing crowding and deep bite on anterior. Consult with patient regarding findings.

An excellent candidate opens by acknowledging Jordan's expectation of a quick referral visit, then pivots to explain that the examination has revealed something important that needs to be addressed first. The candidate explains in plain language that the gum measurements — called CPITN scores — show pocketing of 6 mm or more in all back teeth, and that the X-rays confirm bone loss around the molars. The candidate clearly states that this level of disease in a 17-year-old is not normal and requires investigation beyond a routine scale and clean. The candidate then takes a targeted history: asking specifically about brushing frequency, flossing, and toothbrush type; asking directly and non-judgmentally about cigarette or vape use; asking whether any family members have had gum problems or lost teeth early; and asking specifically whether anyone in the family has diabetes or other systemic conditions. When Jordan discloses his father's early tooth loss and grandmother's

type 2 diabetes, the candidate connects these to a possible familial predisposition to aggressive periodontitis and a potential systemic component. The candidate also asks about stress, sleep, and diet, connecting HSC pressure to immune function and periodontal risk. The candidate articulates a differential that includes Stage III Grade C periodontitis, periodontitis as a manifestation of systemic disease. Management is explained clearly: immediate referral to a periodontist for specialist diagnosis and management; GP referral for fasting glucose and HbA1c given the family history of type 2 diabetes; specific oral hygiene instruction including twice-daily brushing and daily flossing; a brief non-judgmental smoking cessation message; and an explanation that orthodontic treatment must be deferred until the gums are stable, because moving teeth through inflamed bone accelerates bone loss and risks tooth loss. The candidate recommends 3-monthly review and explains why this is more frequent than a standard check-up.

Generalised Periodontitis With Replacement Options **PQ-11**

Periodontics · 11 minutes · Cluster 2

The station. Miss Luiz is a 55 yr old patient, attending your clinic today complaining of pain, bleeding gums and mobility of left lateral incisor (22). OPG given with boneless around most of the teeth and lateral has boneless up to 90%. Perio chart provided with probing depth of 7-8mm around some of the teeth. MH: she is taking thyroxine for hypothyroidism, bone pain and coeliac disease and taking calcium, vitamin D for bone. Patient wanted to know about replacement option.

An excellent candidate would begin by warmly greeting Miss Luiz and confirming her main concerns — pain, bleeding gums, and the wobbly front tooth. They would explain, in plain language, that she has a condition called generalised periodontitis, where bacteria under the gumline have caused the bone that holds her teeth in place to dissolve away over many years. Using the OPG and perio chart, they would show her that most teeth have lost significant bone support, and that the upper left lateral incisor (tooth 22) has lost approximately 90% of its bone, making it non-restorable and requiring extraction. They would acknowledge her medical history — hypothyroidism managed with thyroxine, coeliac disease, and supplementation with calcium and vitamin D — noting these are relevant to healing and bone quality.

. They would outline a staged treatment plan: first, initial periodontal therapy including full-mouth debridement and thorough oral hygiene instruction; second, extraction of hopeless teeth including tooth 22; third, a healing and reassessment phase of at least 8–12 weeks, GP referral for blood tests and bone scan. short term options immediate denture and fiber reinforced bridge

discussion of definitive replacement options.

For replacemen long term, they would present various options — a removable partial denture (most affordable, non-invasive, but less comfortableput pressure on bone and aesthetic) maryland being long term preferred option, a dental implant (most functional and aesthetic long-term solution, but requires adequate bone costs she already has lots of bone loss and poor bone quality so less preferred option for her) approximately \$4,000–6,000 per implant in Australia, with a longer timeline), and no replacement for posterior teeth with an explanation of the drifting and occlusal consequences. They would involve the patient in shared decision-making, address her concerns about appearance and cost, and arrange a follow-up appointment for periodontal therapy.

Chronic Generalised Periodontitis Assessment and Management PQ-17

Periodontics · 11 minutes · Cluster 2

The station. Mr Boldoski is a 55 yrs old patient, attending your clinic today complaining of bleeding gums and wanted some cleaning. You have examined the patient. OPG taken today and you can see generalized bone loss. You gave diagnosis as generalised periodontitis, stage 4 and grade C. His medical records show that he didn't see GP for 10 yrs. You decided to refer him to specialist. Explain the reason for referral and address pts concern.

An excellent candidate would begin by warmly greeting Mr Boldoski and acknowledging his concern about bleeding gums and his desire for cleaning. They would take a focused periodontal history, asking about the duration of bleeding, any pain, tooth mobility, previous dental visits, and current oral hygiene practices. They would then systematically explore medical and social history, specifically asking about smoking and diabetes, and would elicit that Mr Boldoski smokes 10 cigarettes per day with a 30-year history

also linking underlying medical conditions (diabetes/hormonal/ nutritional deficiency) as the patient hasn't done any GP visits in 10 years.

The candidate would explain the diagnosis in plain language: that Mr Boldoski has a serious gum infection called Stage 4 Grade C periodontitis, where bacteria have caused significant bone loss around the teeth — up to 40-50% in some areas — and that some teeth already show mobility. They would explain that this is not simply a matter of 'needing a clean' but a complex condition requiring specialist-level care.

The candidate would clearly explain the reason for specialist referral: the severity of bone loss, multiple deep pockets up to 8mm, tooth mobility on four teeth, vertical bony defects, and the compounding risk factors of uncontrolled diabetes and heavy smoking, which accelerate disease progression and impair healing. They would explain that a periodontist can offer advanced non-surgical and surgical options, including full-mouth debridement, possible osseous surgery, and long-term maintenance planning.

the candidate would strongly encourage Mr Boldoski to see his GP to review his blood work for diabetes nutritional deficiency

Smoking cessation would be addressed empathetically but directly, with a referral to Quitline (13 7848) offered. The candidate would address cost concerns by explaining that early specialist treatment is more cost-effective than eventual tooth loss and implants. They would check understanding throughout and reassure Mr Boldoski that with compliance, tooth loss can be minimised.

When Mr Boldoski raises his travel and cost constraints, the candidate would acknowledge this as a genuine barrier rather than brushing past it, and would mention that options exist to reduce the burden: tele-dentistry or phone review for straightforward follow-up appointments so he isn't always travelling in person, mobile dental clinics that can sometimes bring routine review closer to where he lives, and patient transport assistance schemes that some patients are eligible for to help with the cost of travelling to specialist appointments. The candidate would frame this as "there are supports worth asking about" rather than quoting specific eligibility rules or amounts, and would suggest he raise it directly with the specialist's rooms or his GP when the referral is organised.

if sensitivity is missed in the history then highly likely its management is not addressed (acute care)(not eating due to sensitivity can affect blood sugar level)

Acute Periodontal Abscess PQ-22

Periodontics · 11 minutes · Cluster 2

The station. Jane Highland 55 came to your clinic with sore 11 and 21 with mobility grade II. extruded 11 with 7mm probing depth. generalised 4 to 5 mm probing depth. The patient is fit and healthy, but she smokes 25 cigarettes/day. Manage the case, give the diagnosis, and explain treatment options.

I would begin by introducing myself and establishing rapport before taking a focused history. I would ask Mrs Jane Highlands about the onset, duration, and severity of her pain, noting it started two days ago and worsened yesterday, is throbbing in nature, rates 8/10, and is associated with a bad taste suggesting purulent discharge. I would ask about swelling, fever, difficulty swallowing, and malaise.

Crucially, I would ask about her medical history, specifically her diabetes. Communicates the link between diabetes control and periodontal disease; advises patient to see GP for diabetes review and emphasises importance of compliance with periodontal maintenance and GP visit to review Diabetes status and any deficiencies (explains that unmanaged gum disease can affect glycemic control).

perform vitality testing to confirm the tooth is vital — a critical finding that differentiates a periodontal abscess from a periapical abscess. I would probe the pocket, finding a 7 mm pocket with purulent discharge 11 21. The radiograph confirms 60% bone loss with a vertical defect distally and no periapical pathology.

My primary diagnosis is an acute periodontal abscess on tooth 11 21 with moderate generalised chronic periodontitis Stage III Grade B as the underlying condition. The differential includes periapical abscess, which is excluded by confirmed vitality and absence of periapical radiolucency.

Immediate management involves drainage via the pocket using a curette, irrigation with saline, and if fluctuant, incision and drainage. I would advise continuing paracetamol and ibuprofen.

explains that the diastema and extrusion could self correct after draining and managing gum condition first (and explains further management after review in 1 month if they don't improve)

I would review her in 2 days to assess resolution, on review appointment discuss management of extruded tooth (might go back after we drain the abscess/ trim / refer for ortho) and diastema (goes back after drainage/ filling / ortho) and then schedule generalized periodontal management with specialist therapy including full-mouth debridement under local anaesthesia once the acute phase resolves. I would counsel Mrs Jane Highland on the bidirectional relationship between diabetes and periodontal disease, strongly advise a GP review for diabetes management, and reinforce the importance of 3-monthly periodontal maintenance and daily interdental cleaning and smoking cessation.

Peri-Implantitis PQ-37

Periodontics · 11 minutes · Cluster 3

The station. Mr Dubey 60 year old patient complaining of bleeding gums has Peri-implantitis affecting an implant-supported bridge spanning 32 to 42. A radiograph and clinical picture are provided showing peri-implant marginal bone loss of approximately 4 mm (beyond expected crestal remodelling), with peri-implant redness, bleeding on probing, and probing depths of 6–7 mm. The patient is a smoker; the main concern is the bleeding gums. Explain the diagnosis, treatment options, and management, and address their concerns.

An excellent candidate would begin by introducing themselves and establishing rapport with Mr Dubey, acknowledging his concern about bleeding gums.

They would take a focused history asking about the duration of symptoms, noting redness for 6–8 months and mild chewing discomfort for 2 months, and would directly ask about smoking, eliciting the 10 cigarettes per day history. They would confirm oral hygiene habits, brushing frequency, flossing, and recall attendance.

The candidate would explain recording 7–8 mm depths with bleeding on probing and suppuration, assessing implant mobility (absent), and noting visible plaque accumulation. They would request or review a periapical radiograph confirming approximately 4 mm of bone loss beyond expected crestal remodelling, meeting the threshold for peri-implantitis diagnosis. probing 7 to 8 mm around the implants and 4 mm around other teeth. also explaining the diagnosis of generalised moderate periodontitis

The candidate would explain the diagnosis clearly: peri-implantitis is an infection around the implant causing bone loss, more serious than early gum inflammation around implants (peri-implant mucositis), and is the reason for the bleeding and discomfort. They would distinguish this from mucositis in plain language.

Risk factors would be addressed directly and sensitively: smoking significantly impairs healing and increases bone loss risk(makes the blood flow slow, dry mouth, increases bad bacteria), once-daily brushing without interdental cleaning allows plaque to accumulate around implant components, and irregular recall has allowed the condition to progress undetected. Gout (increased uric acid level) increases inflammation in the gums which increses inflammation in the body lowering kidney function

Management not addressing this would result in loss of implants

short term - phase 1 non surgical therapy above the gum line Scaling (plastic implant scaler/ OH instruction / interdental brush and superfloss use since irregular with GP reviews its a good moment to think and get things under control phase 2 cleaning below the gum line specialist will consider antibiotic (surface decontamination / bone grafting/ implant surface smoothing/ soft tissue grafting)

long term - denture/ bridge

review 3 monthly due to his high risk

The candidate would explain that if non-surgical treatment fails to resolve the infection and halt bone loss, surgical intervention may be required. A review appointment at 8–12 weeks would be arranged, with emphasis on regular 3-monthly maintenance thereafter.

health promotion oral hygiene aids around implants Pixter and super floss need to be regular with GP (Gout) use his dislike for denture and hypertension to motivate him to try quitting linking them to direct result of smoking / quitline 3monthly review

Management of Acute Necrotising Ulcerative Gingivitis PQ-45

Periodontics · 11 minutes · Cluster 2

The station. John Smith, 25-year-old male. Chief Complaint: 'My teeth are bleeding and they hurt. It stings and makes it hard to eat.' Problem has occurred on and off for the past year, gotten worse over the past two months. Went to GP this morning and was given penicillin but hasn't bought it yet. Examination Findings: Ulcers visible around the margins of the gum. Medical/Social History: Medically fit and healthy. Had a viral infection 10 days ago. Smokes 10 cigarettes/day. Takes alcohol once a day. Law student with upcoming exam — stated it was quite stressful. Allergy to penicillin. Clinical Image: Frontal intraoral photo showing classic NUG/ANUG presentation. Task: Explain diagnosis and...

An excellent candidate would begin by introducing themselves and establishing rapport before taking a focused history. They would clarify the onset, duration, and character of pain, ask about recent stress, smoking, diet, sleep, and any systemic symptoms such as fever or lymphadenopathy. Crucially, they would identify the penicillin allergy documented in the brief before any antibiotic discussion. On examination, they would note the pathognomonic punched-out interdental papillae with grey-yellow pseudomembrane, spontaneous gingival bleeding, severe halitosis, and generalised marginal erythema, while confirming the patient is afebrile and systemically well with no lymphadenopathy. They would confidently diagnose Acute Necrotising Ulcerative Gingivitis, explaining to the patient in plain language that this is a bacterial gum infection triggered by the

combination of stress, smoking, poor sleep, and reduced oral hygiene. They would reassure the patient that ANUG is not contagious. Immediate management would include gentle supragingival debridement with warm saline or dilute hydrogen peroxide irrigation, oral hygiene instruction using a soft brush, and advice to avoid smoking. Because the patient has a penicillin allergy, they would prescribe metronidazole 400 mg twice daily (BD) for three to five days rather than amoxicillin, with clear instructions to take with food and avoid alcohol. Appropriate analgesia such as ibuprofen 400 mg with food or paracetamol 1 g if NSAIDs are contraindicated would be recommended. The candidate would advise adequate hydration, nutritious soft foods, and improved sleep. They would arrange review in 24 to 48 hours to assess response, and explain that once the acute phase resolves, definitive periodontal treatment including scaling and root planing will be required. Smoking cessation support would be offered with referral to Quitline 13 7848.

Chronic Generalised Periodontitis Assessment and Management PQ-46

Periodontics · 11 minutes · Cluster 2

The station. *Mrs Wilson, 45-year-old, new patient. Presentation: Attending today requesting a scale and clean — previous dentist of 20 years has recently retired. Medical history clear. Smokes 10 cigarettes/day. Family history of early tooth loss. Has brought along an OPG taken at her last dental appointment (tooth 48 extracted). Examination Findings: Full periodontal examination performed including periodontal charting. Heavy plaque and calculus deposits, with some bleeding on probing at deeper sites. Generalised probing depths of 5–6 mm. Recession of up to 2 mm on the lower anterior teeth. Clinical Image: OPG showing generalised alveolar bone loss, with extraction site at tooth 48. Task: Explain...*

An excellent candidate would begin by warmly welcoming Mrs Wilson and acknowledging that changing dentists after 20 years can feel unsettling. They would take a focused periodontal history, confirming the timeline of any symptoms such as bleeding gums, sensitivity, or loose teeth, and asking directly about her smoking habit and daily oral hygiene routine. After reviewing the OPG and periodontal chart, the candidate would explain the diagnosis in clear lay terms: that Mrs Wilson has a serious form of gum disease called generalised periodontitis, where the bone and supporting structures around most of her teeth have been significantly damaged, and that this goes well beyond what a routine scale and clean can address. The candidate would sensitively link her smoking — ten cigarettes per day — and her family history of early tooth loss as important risk factors that both worsen the disease and reduce the body's ability to heal. They would outline a staged treatment plan: beginning with thorough oral hygiene instruction and full-mouth scaling and root planing under local anaesthesia, followed by a reassessment at six to eight weeks to evaluate the tissue response. They would explain that some teeth may require further specialist periodontal treatment or, if they cannot be saved, extraction. The candidate would strongly emphasise smoking cessation as the single most impactful change Mrs Wilson can make, offering referral to Quitline (13 7848) or her GP for pharmacotherapy support. Prognosis would be discussed honestly but with hope: with compliance, regular three-monthly supportive periodontal therapy, and smoking cessation, disease progression can be halted and teeth retained. The candidate would acknowledge Mrs Wilson's concerns about cost and time, explaining that treatment can be staged to manage both, and that investing in periodontal control now is far less costly than tooth replacement later. Throughout, the candidate would seek Mrs Wilson's understanding and consent at each stage.

Management of Chronic Generalised Periodontitis

PQ-47

Periodontics · 11 minutes · Cluster 2

The station. Mrs Wilson, 45-year-old, new patient. Presentation: Attending today requesting a scale and clean — previous dentist of 20 years has recently retired. Medical history clear. Smokes 10 cigarettes/day. Family history of early tooth loss. Has brought along an OPG taken at her last dental appointment (tooth 48 extracted). Examination Findings: Full periodontal examination performed including periodontal charting. Heavy plaque and calculus deposits, with some bleeding on probing at deeper sites. Generalised probing depths of 5–6 mm. Recession of up to 2 mm on the lower anterior teeth. Clinical Image: OPG showing generalised alveolar bone loss, with extraction site at tooth 48. Task: Explain...

An excellent candidate would begin by warmly introducing themselves and establishing rapport with Mrs Wilson, acknowledging that changing dentists after 20 years can feel unsettling. Before explaining the diagnosis, the candidate would elicit a focused periodontal history — asking about bleeding gums, tooth mobility, previous gum treatment, and current oral hygiene practices including brushing frequency, flossing, and use of interdental aids. Crucially, the candidate would specifically ask about smoking history in pack-years and probe for any systemic conditions, including diabetes, recognising that the examiner context reveals these are hidden risk factors the patient will only disclose if directly asked.

Having gathered this information, the candidate would explain the diagnosis of chronic generalised periodontitis in plain language — describing it as a serious but treatable infection of the supporting structures around the teeth, where bacteria under the gumline have caused the bone and gum to pull away, creating pockets of 5–6 mm. They would reference the OPG to show the generalised bone loss visually and explain that the recession on the lower front teeth is a visible sign of this process.

The candidate would then clearly link smoking and diabetes to worsened disease severity and poorer healing, using non-judgmental, motivational interviewing language to explore readiness to quit smoking and encourage liaison with her GP regarding diabetes control and the bidirectional periodontal-systemic relationship. They would recommend a structured management plan: personalised oral hygiene instruction, full-mouth non-surgical periodontal therapy (scaling and root planing) delivered across multiple appointments, a 6–8 week reassessment, and referral to a periodontist if inadequate response. Cost and time concerns would be addressed honestly, explaining the staged approach and the long-term cost of inaction — progressive tooth loss mirroring her family history. Informed consent would be documented, along with a plan to write to her GP.

Perio Patient — Replacement Options for Missing Teeth (OPG, Bone Loss) PQ-51

Periodontics · 11 minutes · Cluster 1

The station. 45-year-old male, new patient. Has been seeing another dentist. Presenting for a second opinion regarding missing teeth and replacement options. Examination Findings: OPG provided showing generalised bone loss and multiple missing posterior teeth. Patient is known to have significant bone loss. Has not been formally diagnosed at this visit. Task: Gather information. Advise on investigations required. Give appropriate replacement options for missing teeth. No need to provide a formal diagnosis. Focus on patient communication, exploring options (implants, bridges, partial dentures), and realistic expectations given bone loss.

An excellent candidate would open by establishing rapport with David in a direct, no-nonsense manner, acknowledging that he is here for a second opinion and that he deserves straight answers. The candidate would take a focused history covering: duration and frequency of dental attendance, any prior periodontal treatment, compliance with recall, oral hygiene habits including the cessation of flossing, smoking history (ten cigarettes per day for twenty years — a 10 pack-year history), alcohol use, and medical history including a

specific question about diabetes. On identifying that David stopped flossing because his gums bled, the candidate would gently but clearly correct this misconception, explaining that bleeding gums are a sign of inflammation and that stopping interdental cleaning allows the disease to progress unchecked. The candidate would then explain the OPG in plain language: the bone that anchors the teeth has reduced by roughly half in most areas as a result of longstanding gum disease, and several posterior teeth are already missing. The candidate would be honest that this is significant but not hopeless. When discussing replacement options, the candidate would cover all three modalities — implants, fixed bridges, and removable partial dentures — with honest pros and cons. Critically, the candidate would explain that none of these options can be pursued until the active gum disease is brought under control first. Regarding implants specifically, the candidate would confirm that implants are a realistic goal but would explain that smoking approximately doubles the risk of implant failure, that smoking cessation is strongly recommended before proceeding, and that bone grafting may be required in areas of significant resorption. Fixed bridges would be noted as potentially unsuitable given that the neighbouring teeth are themselves periodontally compromised. Removable partial dentures would be presented as the most immediately available option but the least preferred long-term. The candidate would close by reassuring David that options remain available, that this is not a hopeless situation, but that the sequence matters: stabilise the gum disease, re-evaluate, then plan replacements.

Drug-Induced Gingival Enlargement — Cyclosporine + Nifedipine PQ-56

Periodontics · 11 minutes · Cluster 3

The station. *Mr Richard Jones, regular attendee. Chief Complaint: Swollen and bleeding gums. Has a wedding to attend next week — wants it resolved quickly. Examination Findings: Generalised gingival enlargement noted on examination. Periodontal charting performed. Medical History: Kidney transplant — taking cyclosporine for 15 years (immunosuppressant); Hypertension — taking nifedipine (calcium channel blocker) for the past 8 months; Taking atorvastatin for cholesterol. Drug-Induced Gingival Overgrowth: Both cyclosporine and nifedipine are well-known causes. The combination significantly increases risk and severity. Task: Diagnose and manage. Address the patient's concerns. Discuss the drug-gingival...*

An excellent candidate would begin by warmly acknowledging Mr Jones's concern about the wedding and signalling that they will address it directly. Through targeted history-taking, the candidate would establish the precise timeline: gums were stable for 15 years on cyclosporine alone, and the swelling and bleeding began approximately 8 months ago — correlating exactly with the introduction of nifedipine. The candidate would then deliver a clear diagnosis: drug-induced gingival overgrowth, caused by the combination of cyclosporine and nifedipine, both of which independently cause this condition, and together produce a significantly worse effect. The mechanism would be explained in plain language — these medications alter the behaviour of cells in the gum tissue, causing them to produce too much fibrous tissue, and plaque and bacteria make this considerably worse. The candidate would explicitly advise Mr Jones not to stop or alter either medication himself, explaining that stopping nifedipine could cause a dangerous rise in blood pressure, and stopping cyclosporine could risk rejection of his transplanted kidney — both decisions belong with his medical team. The management plan would follow a clear ladder: immediate intensive oral hygiene instruction and full-mouth scaling and root planing, with a review at six to eight weeks to assess tissue response. The candidate would commit to writing a liaison letter to the GP explaining the diagnosis, identifying nifedipine as the likely precipitating agent given the timeline, and requesting consideration of substitution with an antihypertensive with a lower association with gingival overgrowth, such as an ACE inhibitor or ARB. The transplant physician would also be contacted regarding the cyclosporine component and a potential drug interaction. Regarding the wedding, the candidate would be honest: professional cleaning will reduce bleeding and acute inflammation and should produce some visible improvement within a week, but meaningful structural resolution of the overgrowth itself is unlikely in that timeframe and may take several months.

Ortho Referral — Mild Periodontitis (PQ-1 Variation)

NEW

Periodontics · 11 minutes · Cluster 1

The station. *Maya Chen, a 16-year-old Year 11 student, has come to your practice wanting a referral for braces to fix her crowded teeth — her cousin's wedding is in four months and she'd like a plan in place. Last dental visit was 12 months ago. You have done CPITN scoring today (Community Periodontal Index of Treatment Needs), which is 3, 3, 3 / 3, 2, 3. Bitewing X-rays show no significant bone loss. Intraoral picture shows moderate crowding of the lower anteriors with generalised mild gingival inflammation and visible calculus on the lower incisors. Consult with the patient regarding the findings.*

This is the same referral request as PQ-1 — a teenager with crowding who wants braces — but a different periodontal picture, and the correct answer is different too. The candidate who treats every elevated CPITN score in a teenager as a red flag will over-manage this case; the candidate who ignores it will under-manage it. The skill being tested is reading the severity correctly.

CPITN 3, 3, 3 / 3, 2, 3 means five sextants have pockets of 4-5mm, and the lower anterior sextant scores 2 — calculus present, but no true pocketing. That lower-anterior finding is common and unremarkable on its own; that sextant sits opposite the salivary duct openings and accumulates calculus fastest of anywhere in the mouth. This is a world away from PQ-1's 4, 4, 4 / 4, 2, 4 — pockets of 6mm or more in every posterior sextant, which is genuinely abnormal for this age and demands a specialist work-up. Here, a code of 3 still means the affected sextants need detailed periodontal charting today — with five of six elevated, that is effectively full-mouth charting — but code 3 disease is managed with oral hygiene instruction and non-surgical periodontal therapy, not an automatic periodontist referral.

Maya expects a quick referral conversation. Acknowledge that before redirecting: "I can see you're keen to get moving on braces, and we will get there. Before I write the referral, I want to talk you through what I found when I checked your gums today, because it changes the plan a little — in a good way, I think."

Even though this case turns out to be straightforward, take the full history — that is what separates a candidate who got the right answer by luck from one who reasoned their way there:

Put it together: generalised mild gingival inflammation with early, localised periodontal breakdown, driven by inconsistent interdental cleaning that her crowding makes genuinely difficult — not Stage III/Grade C aggressive periodontitis, and not periodontitis as a manifestation of systemic disease. The bitewings support this: no significant bone loss, consistent with early disease rather than established, longstanding attachment loss.

The wedding in four months gives Maya a real but non-clinical deadline. Being honest that a few weeks of perio treatment now still gets her to the orthodontist in plenty of time turns a conversation she might experience as bad news into one she leaves feeling good about.

Treating CPITN 3 the same as CPITN 4 and referring straight to a periodontist. Never mentioning the perio findings and just writing the ortho referral she asked for. Failing to ask why she doesn't floss and missing the crowding-hygiene link entirely. Continuing to probe for a systemic cause after the family history comes back clean. Giving no review interval or escalation pathway. Leaving her with the impression that braces are off the table rather than delayed by weeks.

Uncontrolled Diabetes patient with periodontitis PQ-62

Periodontics · 11 minutes · Cluster 3

The station. *Mrs Lima, 48, a regular attendee, presents complaining of bleeding gums while brushing and sensitivity to cold. Her last dental visit was 6 months ago, when a clean was done and the dentist told her she had mild gingivitis; that dentist used an electric brush to clean her teeth. Today you have examined her, completed periodontal charting, and diagnosed generalised moderate periodontitis; you have also taken an OPG. Her medical records show she is Type II diabetic but is not taking any medication and hasn't been to her GP. She is a smoker, 10 cigarettes/day. Task: manage her concerns and explain your treatment plan.*

The two things this station is really testing are: recognising and explaining the abnormally rapid rate of periodontal breakdown as evidence of uncontrolled diabetes, and not falling into either of the two traps — blaming the previous dentist, or deferring all treatment until her diabetes improves.

OPENING AND HISTORY "Mrs Lima, we've done your examination and I'll explain what we found in a moment, but first, is it okay if I confirm a few things? Can you tell me more about the bleeding and the sensitivity — what tends to bring it on? And could you tell me about your visit six months ago — what was done, was there any numbing injection involved? Have you been fairly regular with dental visits generally?" She'll confirm no injection was given last time — just a clean. Then ask gently about her diabetes: "How has your diabetes been managed since you were diagnosed — are you on any medication, and when did you last have it checked with your GP?" She'll admit she's never started medication and hasn't followed up. Ask about smoking too.

EXPLAINING THE DIAGNOSIS "Normally the space between your gum and tooth is about 2 to 3 millimetres. In your case, some areas are measuring 4 to 5 millimetres, and in places your gum has pulled back, exposing the root — and this is happening in more than 30 percent of your mouth. That means this has moved on from mild gingivitis to what we call generalised moderate periodontitis."

ADDRESSING "DID MY DENTIST DO A BAD JOB? HOW DID THIS HAPPEN SO FAST?" "That's a really fair question, and I want to explain it clearly. Six months ago, a clean without any numbing was exactly the right treatment, because at that point you only had mild gingivitis — there wasn't deep disease under the gum to treat yet. What's happened since is that your diabetes, which hasn't been treated, has driven things to progress much faster than usual. Even in people at higher risk, we'd typically expect around a millimetre of bone loss over a whole year — but it looks like you've lost around three millimetres in just six months, several times faster than expected. That's not a sign anything was missed last time — it's a sign of how much of an effect uncontrolled diabetes is having right now."

THE DIABETES–GUM CONNECTION "There's actually a two-way relationship here. The bacteria involved in gum disease trigger ongoing inflammation in your body, and that inflammation makes it harder for your body to respond to insulin, which makes your blood sugar harder to control. At the same time, when blood sugar runs high, it makes it harder for your gums to heal and fight infection — so each one makes the other worse."

TODAY'S PLAN "Here's what I'd like to do today: I'll give you a clean above the gum line and something to help with the cold sensitivity — either a desensitising toothpaste or a fluoride treatment we can apply here. The bleeding you've noticed is part of the same picture rather than something separate or urgent on its own, so we're treating it as part of this overall plan rather than as an emergency. Because some of these pockets are quite deep and things have moved quickly, I'd like to refer you to a specialist periodontist for the deeper cleaning below the gum line — not because we're waiting for anything, but because a case moving this fast is best managed with their expertise. At the same time, I'd really encourage you to see your GP soon to get your diabetes properly assessed and your blood sugar checked, because getting that under control will make a real difference to how well your gums respond to treatment. I'd also like to see you back here every three months rather than the usual six, given everything going on."

HER ROLE AND HEALTH PROMOTION "There's a lot you can do to help this go well — keeping up good brushing and cleaning between your teeth, coming to your review visits, and getting your diabetes managed will all genuinely improve your chances of keeping your teeth long-term. I also want to mention your smoking — it's connected to how well your gums will heal, and it's worth thinking about for your general health too, not just your mouth. I'll give you something written to take home with everything we've covered."

WHEN SHE ASKS ABOUT COMPLAINTS If she asks how to make a complaint, answer factually and neutrally without being defensive or dismissive, but don't volunteer criticism of the previous dentist — the treatment given six months ago was appropriate for what was present at the time.

COMMON WAYS TO FAIL THIS STATION Telling her nothing can be done until her diabetes is controlled. Criticising the previous dentist or implying something was missed. Giving only a vague "diabetes makes it worse" without the specific rapid-progression comparison. Addressing only sensitivity or only bleeding, not both. Treating the bleeding as requiring urgent same-visit surgical intervention. Not referring to a specialist for the deep component, or not referring her to her GP. Not arranging a shortened recall.

Endodontics

RCT patient wants Extraction PQ-9

Endodontics · 11 minutes · Cluster 1

The station. *Mr Argenti is attending your clinic today and complaining of severe pain coming from one of the tooth on the top left side. You have taken the opg and you can see that some teeth have been Root canal treated few months ago. Patient is very angry and would like to remove these teeth. Give your differentials and explain the necessary investigations and explain extraction option.*

This station looks like an endodontic case and is not one. The pain is non-odontogenic: Mr Argenti has maxillary sinusitis driven by hay fever he has never treated. Everything the station rewards flows from the candidate resisting the pull toward the tooth.

Acknowledge before you assess. Something like: "Mr Argenti, I can hear how frustrating this is — three months of pain, and treatment that hasn't fixed it. I do want to help. Before I can take a tooth out, I need to be sure which tooth is causing this, otherwise I could take one out and leave you in exactly the same pain. Can we work through it together?" That framing does two jobs at once — it validates him, and it pre-justifies the questions he is about to resist.

"Is this pain the same as the pain you had before the root canals, or is it different?" This is the discriminator. Unchanged pain, untouched by definitive endodontic treatment, points away from the tooth. Pain that is new or different in character since the treatment points toward a failed RCT or a vertical root fracture. Here it is the same as before — it never changed — which should immediately widen the differential beyond the teeth.

He will not volunteer the sinus picture. Ask specifically:

Odontogenic: failed or incomplete root canal treatment — a missed canal or persistent bacteria; vertical root fracture. Non-odontogenic: maxillary sinusitis; respiratory infection or allergic rhinitis (hay fever). Two pieces of reasoning worth saying out loud: root canal treatment has a high success rate, so several treated teeth in one quadrant all still hurting should itself make you suspicious of a non-dental cause. Conversely, repeated unilateral sinusitis should raise suspicion of a sinusitis of odontogenic origin — the maxillary posterior root apices sit right against the sinus floor.

"I'm genuinely happy to take a tooth out for you — but only if we can localise it. At the moment several teeth are tender rather than one, and when we numbed that tooth the pain only partly settled. That tells me the pain isn't coming from the tooth itself. If I take it out today, I'd be taking out a healthy tooth and you'd still have the

pain." Respect his autonomy, but do not extract. This is the line the station is testing.

Refer to an endodontist, who will work with ENT to establish the cause — that joint pathway is the expected answer. GP referral for the hay fever and the sinusitis; antibiotics are a GP decision if he develops fever or becomes systemically unwell. Provide interim analgesia advice and book a definite review — he must not leave with nothing after three months of pain.

The untreated hay fever is the single biggest opportunity in this station and the one most candidates miss — he has had it for years and takes nothing for it, and it is plausibly driving the whole picture. Direct him to his GP or pharmacist for management. Reinforce staying off cigarettes (he has already quit — congratulate rather than lecture). Discuss alcohol intake, which also feeds the halitosis. Explain the causes of bad breath, including that a sinus or ear infection can cause it — that will reassure him. Arrange regular dental review.

He will ask whether the last dentist did a bad job, and how to complain. Handle it in three parts:

Agreeing to extract. Never asking whether the pain is the same as before. Taking "no response to cold" on a root-filled tooth as a positive finding. Never asking about the nose. Missing the hay fever entirely, or finding it and doing nothing with it. Criticising the previous dentist. Refusing to engage with the complaint. Ordering investigations without being able to say what each one would tell you.

Patient Flying Interstate PQ-23

Endodontics · 11 minutes · Cluster 2

The station. Mr Fitz Mckenzie is attending your clinic today as an emergency appointment for painful tooth on Lower right back (46). On examination, decayed tooth 46 no gingival swelling, no lymphadenitis and tooth is not tender on percussion. Patient going away for 2 days interstate and has flight in 2 hours, you are also fully booked and can't do any treatment. explain diagnosis, emergency, and definitive management.

An excellent candidate would begin by warmly greeting Mr Mckenzie and establishing rapport before taking a focused pain history. They would ask about the onset, character, duration, and progression of pain — eliciting that it began three days ago as a cold-triggered ache but has evolved into severe spontaneous throbbing pain that wakes him at night. They would ask about aggravating and relieving factors, noting that lying down worsens pain and that cold water previously offered some relief but no longer does.

They would screen for systemic symptoms including facial swelling, fever, and malaise to rule out spreading infection. A brief medical history would confirm no allergies, relevant medications, and contraindications to NSAIDs.

The candidate would then explain the diagnosis clearly: the nerve inside the tooth is severely inflamed and cannot heal on its own — this is called irreversible pulpitis. They would specifically warn Mr Mckenzie about barodontalgia, explaining that pressure changes during the 2.5-hour flight can dramatically worsen dental pain, and that flying with untreated irreversible pulpitis carries a real risk of severe, unmanageable pain mid-flight.

For emergency management, the candidate would strongly recommend performing a pulpectomy — removing the inflamed nerve tissue — today before travel, as this is the most effective way to eliminate the pain source but due to time constraints give local anaesthesia and prescribe pain medicine.

They would explain the procedure briefly and reassuringly. as time does not permit, the candidate would offer to arrange an emergency dental referral in Brisbane if needed, providing contact details for emergency dental services there.

For analgesia, the candidate would prescribe a combination of paracetamol 1g every six hours and Oxycodone 5mg 8 hourly (3days maximum), staggered for continuous coverage, explaining that medication alone is unlikely to fully control irreversible pulpitis pain. They would advise Mr Mckenzie these medicine cannot be have alcohol and driving restrictions.

provide contact for emergency dental services in Brisbane if pain escalates.

The candidate would acknowledge his anxiety and conference commitments with genuine empathy and explores his work life to ease the anxiety.

Root Canal Treatment done by your colleague PQ-24

Endodontics · 11 minutes · Cluster 3

The station. Mrs Burke had RCT done on lower right back tooth 3 days ago with another dentist. Previous dentist didn't explain complications of RCT. Patient has an appointment with a dentist in 2 weeks. She came to see you today because she has pain after the treatment. On examination, band (stainless steel) was given to tooth and GIC filled (temporary restoration). Pain on biting, not sensitive to hot or cold. X-ray provided shows slightly inadequate obturation. give differential diagnosis and manage patients concerns.

An excellent candidate would begin by warmly introducing themselves and establishing rapport, acknowledging that the patient is in pain and that this situation must be distressing. They would take a focused pain history, establishing how much pain is out of 10, throbbing, constant, and worse on biting, and that paracetamol and ibuprofen together have provided only minimal relief.

They would ask about the timeline — that the crack was identified 4 weeks ago after biting hard bread, RCT was completed three days ago, and a temporary restoration with a stainless steel band and GIC was placed. They would screen for signs of infection by asking about fever, facial swelling, discharge from the gum, and trismus, and would note the absence of these features.

Medical history would be taken, identifying well-controlled hypertension managed with perindopril, and confirming no drug allergies. On examination, the candidate would check the temporary restoration for integrity, perform percussion testing (noting marked tenderness on 46), assess mobility (none), palpate the buccal mucosa (slightly tender, no swelling or sinus), check for lymphadenopathy (absent), inspect for the crack line, and critically check the occlusion — identifying a high spot on the temporary restoration.

A periapical radiograph would be requested and interpreted, noting obturation to slightly short in distal canal(could be due to 2d xray/ calcified/ offer to check records) a widened PDL space apically.

The differential diagnosis would be post-endodontic pain or flare up that is likely multifactorial: occlusal trauma from the high temporary restoration, normal post-operative periapical inflammation, and RCT related complications over instrumentation/ irrigant or medicaments pushed beyond apex/ debris bacteria pushed beyond apex/ short filling.

how to explain your colleagues work to the patient and offer to check records. arrange a sit down with your colleague to ease down the situation (1 step of complaint is talk to the dentist or practise manager)

short term Management would include immediate occlusal adjustment of the temporary restoration/ avoid biting from that side, optimisation of analgesia with alternating paracetamol and ibuprofen on a scheduled basis (adding oxycodone if necessary) , monitoring for developing infection.

review

longterm management

The candidate would explain the rationale for RCT in a cracked tooth in plain language and address the patient's concerns empathetically.

health promotion about preventing RCT in the future.

Vertical Root Fracture PQ-41

Endodontics · 11 minutes · Cluster 2

The station. Mr Tiner new patient in your clinic complaining of pain in an upper first molar right side. large MOD composite restoration, and 8 mm pocket in one area of the tooth. you diagnosed it as VRF and decided that the tooth should be extracted he has history of asthma and arthritis taking Ventolin and prednisolone 15 mg explain the diagnosis and management to the patient

An excellent candidate would begin by introducing themselves and establishing rapport with Mr Tiner before taking a focused history.

They would ask about the onset, duration, and character of the pain, any previous dental treatment on the tooth, and specifically whether there has been any trauma. They would elicit the history of root canal treatment completed three years ago, and the patient's avoidance of chewing on the left side.

The candidate would explain a diagnosis of vertical root fracture, explaining that the combination of a root-canal-treated tooth, isolated deep narrow pockets.

They would communicate clearly and empathetically to Mr Tiner that the tooth cannot be saved and requires extraction.(can consult with the specialist if hemisection can be done)

risk factors - diet/ grinding/ RCT/ Large filling

explain how it happened

Before proceeding, the candidate would note the patient's medical history —he has history of asthma and arthritis taking Ventolin and prednisolone 15 mg for 4 weeks. as it affects the management

risk of adrenal crisis? (adrenal gland wont produce stress hormone in patient taking prednisolone 20mg for more than 2 weeks resulting in drop in sugar level and blood pressure due to adrenal supression

Management emphasising the importance of pain management as the tooth cannot be removed today as the adrenal crisis risk guidelines are not met and focus on pain

-today xray to decide if i can remove this tooth

- temporise grinding and mild pain band / pain medicine paracetamol + oxycodone / occlusal stops (antibiotic if infected.

plan tooth removal (action plan/ accompanied by / adult / morning appointment) asthma inhaler present .

hospital referral if theres severe pain

-long term replacement options / splint

health promotion about hard diet/ grinding and regular review

Consent for Endodontic Treatment PQ-48

Endodontics · 11 minutes · Cluster 3

The station. Mr Jonathan, 27-year-old, regular patient. Scenario: You saw him in the last visit and gave the diagnosis of irreversible pulpitis for tooth 16. Patient is back again complaining about pain and does not want extraction. Medical History: Medically fit and healthy. Doesn't drink or smoke. Clinical Image: Periapical radiograph showing tooth 16 (upper first molar) with roots in close proximity to the maxillary sinus. Task: Explain the procedure and gain informed consent for endodontic treatment (roots close to sinus).

The candidate begins by confirming the patient's identity and the tooth requiring treatment, clarifying that today's appointment is to discuss root canal treatment for tooth 16, the upper right first molar. They explain the

diagnosis in plain language: the nerve inside the tooth has become severely inflamed and cannot heal on its own, which is why the pain has been persistent and worsening. They clarify that a filling alone will not resolve this because the problem lies deep inside the tooth's nerve, not just the outer surface.

The candidate then describes the root canal procedure in accessible terms: the tooth is numbed with local anaesthetic, the inflamed nerve tissue is removed, the canals are carefully cleaned, shaped, and disinfected, and then sealed with a filling material. A crown or permanent restoration is placed afterwards to protect the tooth. They note that treatment typically requires two to three appointments.

Benefits are clearly outlined: relief of pain, preservation of the natural tooth, and avoidance of extraction. The candidate then addresses material risks specific to this case, including the proximity of the roots to the maxillary sinus — explaining that there is a small risk of communication with the sinus during treatment, which would be managed appropriately. Other risks discussed include treatment failure (approximately 5–10%), persistent or new pain, instrument separation within the canal, root perforation, tooth fracture, and the possible need for retreatment or a minor surgical procedure called an apicectomy.

Alternatives are presented: extraction with or without replacement via implant, bridge, or denture, or no treatment with the risk of spreading infection and worsening pain. Pain management is addressed directly, reassuring the patient that modern root canal treatment is performed under effective local anaesthesia and is generally comfortable, with over-the-counter analgesia such as paracetamol or ibuprofen recommended post-operatively. The candidate checks understanding, invites questions, and confirms the patient's willingness to proceed.

Restorative dentistry

Peg Lateral Incisor — Explain and Plan PQ-10

Restorative dentistry · 11 minutes · Cluster 2

The station. Miss Nguyen is a new patient at your clinic, attending today complaining about the appearance of her front tooth. She works in the pub and her partner thinks she has vampire tooth. MH; patient has anxiety in general and taking diazepam 10mg twice/day. Diagnose and manage patient concern.

An excellent candidate would begin by warmly greeting Miss Nguyen, establishing rapport, and inviting her to describe her concern in her own words. They would acknowledge her self-consciousness and validate that this is a common and very treatable condition. After confirming her medical history — noting her diazepam use for anxiety and considering its implications for treatment planning and sedation — the candidate would explain in plain language that her 'vampire tooth' appearance is caused by a peg lateral incisor, a developmental variation where the tooth simply grew smaller than usual, and that this is not a sign of disease or poor oral hygiene. The candidate would clarify that both upper lateral incisors appear to be affected, which is actually common when this occurs. They would then outline three main treatment pathways: first, composite resin build-up — a conservative, tooth-friendly option where tooth-coloured filling material is shaped around the existing tooth to create a normal appearance, requiring little to no drilling, fully reversible, and costing approximately \$200–\$400 per tooth; second, porcelain veneers — thin ceramic shells bonded to the front surface, requiring minimal but irreversible enamel reduction, offering superior aesthetics and longevity of 10–15 years, at approximately \$1,500–\$2,500 per tooth; and third, full crowns — generally not recommended here as they require significant tooth reduction and are disproportionate to the clinical need. The candidate would address her concern about damaging her teeth directly, reassuring her that composite build-up preserves virtually all natural tooth structure.

They would note that her private health insurance may contribute to composite restorations. Given her anxiety, the candidate would adopt a calm, unhurried approach, offer written information, and involve her in shared

decision-making by recommending composite build-up as a reversible starting point, with the option to upgrade to veneers later. A follow-up treatment planning appointment would be offered.

broucher explaining treatment options also seeking help from someone close to decide the treatment option and treatment descion should come from her as shes seeking treatment after a comment from her boyfriend who shes dating since 3 months(domestic violence of dominating nature can be suspected here but not sppoken directly about)

Amalgam Filling done by a colleague PQ-30

Restorative dentistry · 11 minutes · Cluster 3

The station. *Scarlette Jones is a regular patient in your clinic. She is attending today complaining of severe pain related to one of her upper left first molar tooth. She had an amalgam filling done by one of your colleagues 2 weeks ago while you were away which chipped off 2 days after it was done. She is very upset, and she hasn't been able to sleep for the past 2 nights. You took an Xray and found out that there are some caries under amalgam filling and apical radiolucency associated with that tooth and because of which she is experiencing pain. How will you manage her concerns and give her appropriate treatment options?*

An excellent candidate would begin by warmly acknowledging Scarlette's distress and validating her experience — two sleepless nights with pain is genuinely difficult — before taking a structured pain history. They would ask about the onset, character (worse on biting), duration of episodes, and specific triggers including cold, heat, and chewing, and would confirm how she has managed the pain so far and what analgesia she has already tried. Before proceeding, they would confirm her medical history, current medications, and any allergies. They would ask specifically how the pain started, whether she noticed anything odd — a high spot or discomfort — immediately after the filling was placed, and how the restoration came to chip. The candidate would explain the diagnosis of acute apical periodontitis in plain language: that the nerve inside the tooth had likely already died before or around the time the filling was placed, and that this created a pocket where gas and pressure could build up, which is what is causing her pain now; they would note that a radiographic shadow at the root tip typically takes a minimum of three weeks to develop, meaning the underlying infection was probably already present at the time of the original filling rather than being caused by it. They would ask whether her colleague had taken a radiograph before placing the filling — establishing that the patient recalls none being taken — and would explain diplomatically that this may have been because the tooth was painless at the time, even though a preoperative radiograph is recommended practice before a filling of this depth. For interim management, they would perform a pulpectomy and place a medicated dressing to relieve the acute infection, and would recommend appropriate analgesia such as ibuprofen 400mg alternated with paracetamol 500–1000mg. They would then present two definitive treatment options for the longer term: root canal treatment to save the tooth — noting that given how heavily restored it is and the narrow canals visible on the radiograph, referral to a specialist would be preferable to treatment in-house — explaining the likely number of appointments, the eventual need for a crown, and the associated costs; or extraction, which they would offer without further cost, adjusted against what was already paid for the filling, discussing the advantages and disadvantages of each option honestly. For the longer term, they would discuss a crown following root canal treatment, or replacement options such as an implant or bridge should extraction be chosen instead. They would provide health promotion advice around the importance of regular review, and regarding the colleague, would offer to arrange a sit-down conversation and explain the practice's complaints process clearly and neutrally. They would document all findings and the discussion contemporaneously.

Amalgam Replacement Discussion PQ-35

Restorative dentistry · 11 minutes · Cluster 2

The station. Mr Chen a 55-year-old medically fit patient, attending your clinic today concerned about mercury poisoning due to his amalgam filling that has been present in his mouth for the past 10 years by his previous dentist. He wants to replace all his amalgam fillings with composite. There is no fracture of restoration, pain, or sensitivity. Explain how would you manage this patient and how you plan to address his concerns?

An excellent candidate would begin by warmly welcoming Mr Chen and acknowledging his concerns without dismissal, recognising that anxiety about amalgam safety is understandable given how much information circulates on social media. They would ask him to share what he has read or been told, and would explore whether he is experiencing any symptoms he attributes to his fillings. After listening carefully, they would explain that the current Australian evidence-based position — supported by the NHMRC and the Australian Dental Association — is that dental amalgam is safe and effective for the vast majority of patients. They would clarify that the mercury in amalgam exists in a stable, bound form and is not equivalent to the organic methylmercury associated with toxicity, and that his fillings, being ten years old and well-established, pose minimal ongoing risk, since the greatest release of mercury vapour actually occurs during placement and removal rather than during everyday function. They would explain the risks of unnecessary replacement: irreversible loss of healthy tooth structure, the possibility of pulpal irritation or damage, the risk of the new restoration itself failing, and the financial cost, noting that composite restorations in load-bearing posterior sites generally have a shorter average lifespan than well-maintained amalgam. They would identify the situations where replacement genuinely is indicated — fracture, secondary caries, poor marginal integrity, a documented mercury allergy, or aesthetic concerns in visible areas — and would note that, based on his history, none of these currently apply. They would ask about any history of teeth grinding, since this would significantly affect the choice and durability of any replacement material chosen. If removal did go ahead, they would explain that it would be carried out using the SMART safe mercury and amalgam removal protocol, with rubber dam isolation and high-volume suction to protect both Mr Chen and clinical staff, and with disposal of the amalgam waste according to Australian guidelines. They would discuss composite resin as a viable alternative material, being transparent about both its aesthetic advantages and its limitations in high-stress posterior sites. Ultimately, they would reach a shared decision with Mr Chen — recommending monitoring and regular review rather than immediate replacement — and would provide him with reliable resources such as ADA patient information and the NHMRC guidelines, inviting him to return with any further questions. If relevant details about his lifestyle emerge during the consultation, such as smoking, the candidate would use the opportunity sensitively to raise smoking cessation as a further general health benefit, separate from the amalgam discussion. They would document the discussion and his informed decision.

International Student with large silver filling PQ-38

Restorative dentistry · 11 minutes · Cluster 1

The station. Mr Gamage, 45-year-old overseas student, came to Australia 2–3 years ago. Chief Complaint: pain from one of his teeth on the lower right side for the past 2 weeks. He cannot localise a particular tooth, but biting on that side makes it worse. Examination Findings: large MOD amalgam restoration on 46, placed around 20 years ago. Clinical Image: intraoral photo showing a large, aged MOD amalgam restoration on tooth 46 with visible fracture lines running through the amalgam. Task: gather information, explain the investigations you would carry out, give a possible diagnosis, and address the patient's concerns.

A strong candidate recognises early that this is not simply a failing filling. They take a focused pain history — two weeks, lower right, cannot be localised, worse on biting — and then go looking for the mechanism rather than stopping there. Asking about diet, or about exactly when the pain began, elicits that Mr Gamage was biting on ice when the tooth cracked.

They then ask about waking symptoms, which is where the station really opens up: sore muscles and headaches on waking, sometimes. A candidate who never asks this misses the entire TMJ and bruxism thread that the case is built around.

Getting to the cause takes persistence. Mr Gamage is cranky, tired and working night shifts as a security guard with pub work on top of study, and he will not hand over the stress on a single closed question. The candidate asks openly, gives him room, and comes back to it — reaching the burnout picture underneath rather than accepting the first deflection.

On investigations they are specific rather than vague: muscle palpation, a bite test, an IOPA or bitewing, and transillumination to look for the crack. When Mr Gamage asks directly how anyone would know whether the tooth has actually cracked, they answer him plainly instead of deferring. They work towards a clear view of whether the pulpitis is reversible or irreversible rather than leaving pulpal status undefined.

Management is staged. In the short term: a custom-made full-coverage splint, and removal of the existing filling with temporisation. In the longer term: a crown, or root canal treatment followed by a crown, depending on the pulpal diagnosis reached.

Finally, they treat the working pattern as part of the problem, not background colour. Health promotion around stress management and preventing burnout belongs in the answer, because the tooth is the presenting complaint and the way Mr Gamage is living is the reason it presented. Throughout, they stay patient with a man who is short-tempered because he is exhausted and sore, rather than matching his tone.

A candidate performing at the top of the band also handles the questions Mr Gamage puts back to them — why the pain comes on releasing rather than biting down (the release causes a snap-back across the crack), how a crack would actually be identified, whether it would show on an X-ray, and how he would even know whether he grinds. They answer these directly rather than deferring, and they use the fact that he cannot localise the pain as positive diagnostic information: poorly localised pain points away from apical periodontitis and towards a cracked tooth, since proprioception would let him pinpoint the tooth if the periapex were involved.

Five-Year-Old with abscess (possibility of Molar Incisor Hypomineralisation) PQ-39

Paediatric dentistry · 11 minutes · Cluster 1

The station. Alice Zang, 5-year-old child, attending with her mother for the first time. Chief Complaint: Pain from a tooth on the lower left side. Dad mentions she couldn't sleep last night. Examination Findings: Second primary molar on the left side has large decay with pain on biting in that area. She took antibiotics when she was under the age of 1.5 years due to an ear infection. Clinical Image: Intraoral photo of lower left molar region showing a decayed primary molar. Task: Gather information, address patient's concern.

An excellent candidate would begin by warmly greeting Alice and her mother, acknowledging the difficult, sleepless night they have had, before taking a focused history. They would ask about the onset and duration of the pain, its severity, any systemic symptoms including fever and malaise, her eating and drinking status, and any previous dental experiences. On reviewing the clinical findings, the candidate would systematically identify an acute dentoalveolar abscess associated with the lower left second primary molar. They would explain the diagnosis to Alice's parent in plain language — that Alice has a dental abscess, meaning an infection has spread from inside the tooth into the surrounding bone and gum, and that the tooth itself is at high risk from the extent of the decay present. They would explore the relevant risk factors — oral hygiene, the regularity of dental review, and diet, including the role saliva plays in protecting teeth — and would note that Alice's antibiotic use for an ear infection before eighteen months of age falls within the critical window for enamel development, raising the possibility that some of her later teeth may show Molar Incisor Hypomineralisation as

a result. For investigations, they would describe an extraoral examination for fever, swelling, lymph node involvement, and jaw joint tenderness, an intraoral examination, a periapical radiograph to assess whether the tooth can be saved, and bitewings to check the remaining teeth, along with a diet history using a three-day diary covering two weekdays and one weekend day. If the family raised financial concerns, they would check eligibility for the Child Dental Benefits Schedule. For treatment, they would explain that the decision to save or extract the tooth depends on Alice's cooperation: if she is able to cooperate, the tooth would be temporised with an odontopaste dressing and referred for root canal treatment with a specialist; if she is not, amoxicillin syrup would be prescribed alongside referral for definitive management. They would provide health promotion advice covering supervised brushing, ideally with an electric toothbrush, and dietary guidance, and would outline a long-term preventive plan for any teeth affected by Molar Incisor Hypomineralisation, including fluoride varnish, fissure sealants, desensitising agents, and regular review, given the likely link to her early antibiotic exposure. The candidate would close with clear safety-netting advice: to present to the emergency department if Alice develops difficulty breathing or swallowing, rapidly worsening swelling, a high fever, or appears generally unwell.

Assessment and Management of Dental Fluorosis

PQ-40

Restorative dentistry · 11 minutes · Cluster 2

The station. Kim, 15-year-old girl. Chief Complaint: Complaining about white and brown patches on all of her teeth. Examination Findings: White and brown spots throughout the mouth. History: Lived in a rural area until she was 10, then moved to boarding school. Took fluoride tablets until age of 10. Doesn't want any drilling or local anaesthesia. Wants teeth fixed ASAP — has a photoshoot at sister's wedding in a few weeks. Clinical Image: Frontal photo of upper and lower teeth showing generalised white and brown/yellow mottled staining across all visible teeth — classic dental fluorosis presentation. Explain diagnosis, treatment planning and management.

An excellent candidate would begin by warmly greeting Kim understand her concerns

then take a focused history exploring fluoride exposure during the critical period of tooth development — birth to age eight. They would ask about the water source in the rural area where Kim lived until age ten, whether fluoride tablets were taken alongside fluoridated water, and whether she swallowed toothpaste as a young child. They would also ask about any childhood illnesses, fevers, medications such as tetracyclines, or dental trauma to differentiate fluorosis from other enamel defects such as molar-incisor hypomineralisation or amelogenesis imperfecta.

The candidate would then explain the diagnosis clearly and sensitively: the white and brown patches are dental fluorosis, caused by excess fluoride during enamel formation in early childhood (1500 to 2000 ppm fluoride before the age of 6 when the enamel of the teeth was forming results in subsurface porosities due to hypomineralization). They would reassure Kim that the enamel is structurally sound, the teeth are not weaker, and there is no increased risk of decay. The candidate would acknowledge that multiple fluoride sources — rural water, fluoride tablets, and toothpaste — likely contributed, and would avoid assigning blame, particularly reassuring the parent that this was not a result of negligence.

Addressing Kim's immediate concern about the wedding photoshoot, the candidate would explain that several options exist

For her mild fluorosis with intact enamel, microabrasion can reduce the brown spots. Resin infiltration (Icon) is a minimally invasive option that masks white spot lesions without drilling or anaesthesia — directly addressing Kim's preference. Diet berry juices can stain will need polishing visits

microabrasion with bleaching for even tone of teeth (bleaching need to take advice from the specialist)

Composite veneers or porcelain veneers/ no prep veneers (bulky but no drilling and numbing needed) would be deferred until adulthood given her age. The candidate would involve Kim in the decision, explain realistic timeframes, and note that resin infiltration could potentially be completed before the wedding. They would also recommend monitoring and reassess at eighteen for more definitive options if desired.

exploring how she'll be paying for the treatment and making her understand that we need consent from the parent paying if paying from their card (important for record keeping)

health promotion fluoride toothpaste / G C tooth mousse to put minerals back Boucher explaining all options and making her understand the importance of involving someone in the decision making as this is cosmetic and some procedures carry a risk of damage to the teeth offering to contact parents or legal guardians

Vague Pain on the Left Side — Interproximal Caries & Open Contact (26/27) PQ-52

Restorative dentistry · 11 minutes · Cluster 2

The station. Female, new patient attending for routine check-up and clean. Chief Complaint: Food lodgement between teeth 26 and 27. Vague pain/aching sensation on the upper left side, mentioned at last appointment with previous dentist. Examination Findings: Radiograph provided showing: radiolucency involving the contact point between 26 and 27 (interproximal caries extending into dentine), affecting the marginal bone; Gap/open contact between 26 and 27; Temperature testing: 35°C response noted; Generalised amalgam restorations present. Task: Carry out history taking and examination. Determine investigations. Establish a working diagnosis and explain findings to the patient.

An excellent candidate begins by warmly welcoming Maria as a new patient and establishing rapport before taking a focused history. They ask open-ended questions about the vague upper left aching — eliciting onset (approximately eight months ago, though closer to eighteen months on gentle probing), character (dull, aching rather than sharp or spontaneous), triggers (cold, sweet foods), and critically, whether the pain lingers after the stimulus is removed. They specifically ask about food packing between the upper left back teeth, confirming the chief complaint. The candidate then asks directly about parafunctional habits — grinding, clenching, or morning jaw soreness — eliciting Maria's disclosure that her partner has noticed nocturnal grinding and that she wakes with a sore jaw. This is integrated into the management plan. On examining the bitewing radiograph, the candidate reads it systematically: confirming image quality, correct angulation, and patient identification before describing findings. They identify a well-defined radiolucency at the distal surface of 26 extending through enamel into dentine, an open contact between 26 and 27, and subtle loss of crestal bone height at the interproximal crest. They note the existing amalgam restorations appear radiographically intact, explaining why the lesion was not visible clinically. The candidate correlates the prolonged thermal response at 35°C with reversible pulpitis, clearly distinguishing this from irreversible pulpitis, while acknowledging that monitoring is essential and root canal treatment for 26 cannot yet be excluded. When Maria asks why her previous dentist missed this, the candidate validates her concern empathetically — explaining that interproximal caries hidden beneath a contact point is invisible to clinical examination alone and that bitewing radiographs are specifically required to detect such lesions — without criticising the previous practitioner. Management proposed includes restoration of the distal surface of 26 with re-establishment of the contact point, periodontal assessment of the 26/27 region, pulp vitality monitoring with a review appointment, an occlusal splint given the bruxism history, and a six-month recall bitewing radiograph.

16-Year-Old with Rampant Caries & Erosion — Dietary + Lifestyle Causes PQ-57

Restorative dentistry · 11 minutes · Cluster 3

The station. Sam, 16-year-old boy, attending with his father. Chief Complaint: Sensitivity to cold on several teeth. Father is accompanying him. History: Regular patient — seen 1 year ago, teeth were ok at that time; Started a new part-time job as an apprentice approximately 1 year ago; High sugar intake; drinks a lot of soft drink during work; Dietary change coincides with onset of sensitivity. Examination Findings / Clinical Images: Cervical caries on tooth 12 and white spot lesions (early caries) on anterior teeth; White chalky patches (enamel demineralisation) on multiple teeth; Caries at cervical margins of multiple posterior teeth, also affecting the marginal bone in some areas; Erosion pattern...

An excellent candidate would begin by warmly greeting both Sam and his father, then immediately establish Sam as the primary patient by directing the first question to him: 'Sam, can you tell me in your own words what's been bothering you?' The candidate would acknowledge Mr Tran's presence and concern early, giving him a defined supportive role while consistently redirecting clinical questions to Sam. History-taking would systematically quantify acidic drink consumption — asking specifically about energy drinks and soft drinks at work, sports drinks after exercise, and what Sam drinks at home versus on-site. When Sam initially underreports, the candidate would probe gently: 'What do you and your workmates usually grab during breaks?' to elicit the true two-to-four cans per day figure. Crucially, the candidate would ask directly: 'Do you ever brush your teeth at work, or after having a drink?' — uncovering Sam's habit of brushing immediately after acid exposure and correcting it clearly: 'Brushing straight after an acidic drink actually scrubs away softened enamel — we need to wait at least 30 to 60 minutes first.' The candidate would rule out GORD and eating disorders as alternative acid sources. Clinical findings would be explained by distinguishing caries — a bacterial process where mouth bacteria ferment sugar and produce acid — from erosion, a direct chemical dissolving of enamel by dietary acid, independent of bacteria. Both are present and both are being driven by the same drinks. The twelve-month trajectory would be used motivationally: 'One year ago your teeth were fine. In twelve months you've developed multiple early lesions and one cavity — if this continues, the damage in the next year will be even greater.' The management plan would include same-day fluoride varnish, prescription of Colgate Neutrafluor 5000 ppm toothpaste, advice to replace between-meal drinks with fluoridated tap water, fluoride mouthrinse at a separate time, and restorations for teeth 12 and 26. White spot lesions would be monitored conservatively. A three-month recall would be scheduled.

Prosthodontics

Management of Denture Stomatitis in a Diabetic Patient PQ-3

Prosthodontics · 11 minutes · Cluster 2

The station. Mrs Smart 65 years old, wearing upper denture for 20 years. Current denture is 20 years old. She complains of broken denture. Denture falls during yawning and eating. She has few teeth on the lower jaw but doesn't wear any denture on lower. On examination you found red inflamed palate, upper denture has buccal flange overextension, all the teeth worn out. Consult with the pt about the management.

An excellent candidate would begin by warmly introducing themselves and establishing rapport before taking a focused history. They would ask about the duration and fit of the current denture, overnight wear habits, and the patient's denture cleaning routine — specifically whether she soaks her dentures or uses denture cleaner. They would then explore her medical history, asking directly about her diabetes, how it is managed, and her most recent HbA1c result. They would also ask about dry mouth symptoms and any medications that might contribute to xerostomia, as well as dietary habits such as sugar intake.

Based on the history and clinical findings — diffuse palatal erythema beneath a poorly fitting, plaque-laden upper denture in a patient with poorly controlled Type 2 diabetes who wears her denture overnight — the candidate would diagnose Denture stomatitis, most likely candidal in origin. They would clearly explain to Mrs Smart that the redness and soreness are caused by a fungal infection called Candida, and that several factors are working together to cause this: her denture acting as a reservoir for the fungus, wearing it overnight preventing the palate from recovering, infrequent cleaning allowing plaque and organisms to accumulate, reduced saliva flow reducing natural oral defences, and poorly controlled blood sugar providing an environment in which Candida thrives.

Management for ill fitting denture offered as chairside or lab relining explaining the advantages and disadvantages of both

Management would be explained in clear stages. First, she must stop wearing her denture when partner is not around wash it with soap and water Denture stomatitis is something you have noticed it's not a presenting complain don't jump to denture stomatitis without addressing their concerns first also explain why they should delay the denture as she has denture stomatitis and can affect the fit of denture. She should brush the denture with a soft brush and rinse thoroughly. Use a towel underneath to prevent breakage in future

If they don't like soap offer denture cleaning tablets

Second, the candidate would review after 2 days and if the diabetes is unstable then this requires specialist advised (candida infection in immunocompromised patient requires aggressive treatment and they may prescribe systemic anti fungal) also if diabetes is normal review after 1 month if there's no improvement Prescribe Anti fungal gel,

Angular Chelitis is an extension of what is happening inside our mouth. Management includes numbing gel swab test to confirm if it due to candida (anti fungal cream) blood test to confirm if it is due to deficiency.

Third, they would address the overextended buccal flange and poor retention, explaining that a new denture will be needed once the infection resolves and her diabetes is better controlled. Finally, they would recommend she see her GP to review her HbA1c, explaining that improving blood sugar control is essential for the infection to resolve and for any future dental treatment to succeed.

Replacement Options After a Posterior Extraction

PQ-7

Prosthodontics · 11 minutes · Cluster 3

The station. 70 years old lady, a new patient in your practice, is a regular dental attendee. You have done some scaling and taken an x-ray; you inform the patient they don't have any new cavities and her gums are healthy. Then she casually asks what replacement options are there for the gap between 34 and 36 (missing 35) present between her teeth. Intraoral picture and x-ray of the left side given showing the gap is small — implant suitability will need to be confirmed with further imaging (e.g. CBCT). On x-ray, 36 has an under-filled RCT and a metallic cap. Management and replacement options.

An excellent candidate would begin by warmly acknowledging the patient's question and taking a brief focused history before launching into options. They would confirm that tooth 35 was extracted four weeks ago due to a vertical root fracture, that healing has been uneventful, and that the patient has no pain or swelling. They would note the adjacent teeth 35 and 37 are minimally restored, the patient's well-controlled hypertension on Perindopril, and her concerns about cost, appearance, and function. The candidate would then explain that doing nothing is an option but carries real consequences — the opposing tooth (45) may over-erupt, and teeth 34 and 36 may drift or tilt into the space over time, compromising chewing and making future replacement harder. They would present three main replacement options. First, a dental implant: a titanium fixture placed into the jaw bone after adequate healing (typically three to six months post-extraction), with a crown placed on

top. It does not involve preparing adjacent teeth, has the best long-term prognosis (15-plus years with good care), but involves surgery, a longer treatment timeline, and the highest cost (approximately \$4,000–\$6,000 out of pocket in Australia). Second, a three-unit fixed bridge using teeth 34 and 36 as abutments: a fixed, non-removable option that restores function and aesthetics well, is completed in two to three appointments, and is less costly than an implant, but requires irreversible preparation of two otherwise sound teeth and demands careful flossing technique underneath. Third, a removable partial denture: the most affordable and least invasive option, but removable, potentially bulky, and often the least preferred by patients long-term. The candidate would acknowledge the patient's reluctance toward a denture, validate her preference for something fixed, and recommend a CBCT or OPG assessment to confirm bone volume for implant planning. They would invite questions and summarise next steps clearly.

Patient wants implants (Heart Valve) PQ-27

Prosthodontics · 11 minutes · Cluster 1

The station. John Edward in a new patient in your clinic want to replace his back teeth with implants patient had a prosthetic heart valve replacement surgery 2 yeras ago and takes regular medicine for it. Gather information and explain necessary investigations. and address patients concerns.

An excellent candidate would begin by warmly introducing themselves and confirming John's chief concern — replacing his missing back teeth with implants. They would take a structured medical history, starting with open questions before moving to targeted enquiry. On identifying the prosthetic heart valve, they would ask about the specific anticoagulant medication — recognising that 'two little pills' likely represents warfarin — and explain that an INR check will be needed before any surgical procedure, and would also note his recently diagnosed hypertension as relevant to bleeding risk and dry mouth. Before discussing the implant procedure itself, the candidate would address the key factors that determine suitability: the presence of any abnormal occlusal forces, the health of his gums, and the amount of bone remaining at the sites. They would explain the patient-related risk factors in plain language — an elevated risk of peri-implantitis given his history of gum disease, excessive forces now that his bite has shifted onto his front teeth, generalised tooth wear, and bone quality and quantity concerns given the proximity of the maxillary sinus above and the nerve below, his history as an ex-smoker, how long ago the teeth were lost, and the forces his current denture places on the bone — noting that a bone augmentation procedure by an oral surgeon may be needed if bone volume is insufficient. They would then explain the procedure-related risks tied to his medical history: the need for antibiotic cover given his prosthetic heart valve, careful management of his warfarin and INR, the effect of alcohol on both bleeding and bruxism, and the effect of hypertension on both bleeding and dry mouth, framing the overall concern as one of avoiding bacteraemia and the small but serious risk of infective endocarditis. For investigations, they would describe a thorough intraoral and extraoral examination — assessing the bite, muscle activity, face height, and remaining teeth, a Basic Periodontal Examination to assess gum health, and an OPG as a general survey — explaining that these findings would be shared with a specialist, who would arrange a CBCT to plan the case in detail. They would outline the implant process in lay terms, present a removable partial denture as an alternative, invite John's preferences, and check his understanding throughout using teach-back. They would close by summarising the next steps and offering relevant health promotion advice.

Flabby Ridge — New Denture for a 20-Year-Old

Loose RPD NEW

Prosthodontics · 11 minutes · Cluster 3

The station. Mr Warren James is here today looking for a new denture, as his current RPD is 20 years old and loose. On examination, in the upper jaw only two teeth are present — 17 and 27, both grade 2 mobile — and the upper ridge is flabby. The lower arch has all natural teeth (photograph given, front view). He is medically fit and healthy. Address and manage the patient's concerns.

An excellent candidate starts by taking Mr James's request seriously — he has come in for a new denture and should not be told bluntly that he cannot have one. They establish how the current denture is failing and, importantly, why he is attending now: the teeth are no longer visible in the worn old denture, so this is as much an aesthetic complaint as a functional one.

They take a history that uncovers the accident as the reason for the tooth loss, and they do not accept "medically fit and healthy" as the end of the medical history — they explore the blood test six months ago and the multivitamins he is taking.

On examination they identify the central problem and, crucially, explain its cause: the upper ridge has become flabby because of excessive forces from the full complement of natural lower teeth acting on the upper anterior area. They state the diagnosis of flabby ridge explicitly, and give a prognosis for the two remaining upper teeth, 17 and 27, both grade 2 mobile. They notice, or ask about, the angular cheilitis and the skin fold at the corner of the mouth, and connect this to a loss of vertical dimension rather than treating it as an incidental finding.

The core of the answer is an honest explanation delivered without alarm: they are happy to make a new denture, but it will not be straightforward. If a new denture is simply built like the old one with clasps on 17 and 27, the flabby anterior ridge means tipping forces at the back will, in time, cost him those two remaining teeth. The better option is to remove them and provide a complete denture, which has a better prospect of success. They can discuss the impression technique a flabby ridge requires, rather than treating it as a routine impression.

They set out the longer view: an implant-supported overdenture or bridgework is the recommendation, because the forces at play will continue to resorb bone, and the longer this is left the more difficult those options become. A transitional denture can be provided in the meantime.

On referral they are specific. This goes to a prosthodontist — the specialty that handles complex removable and implant prosthodontics, which is exactly what a flabby ridge, a selective-pressure impression and an implant-supported overdenture require. If implants proceed, placement may involve an oral and maxillofacial surgeon or a periodontist, with the prosthodontist restoring. Crucially, they frame the referral as the right first call rather than something to try after a conventional denture has failed: the situation is already compromised, and attempting a straightforward remake risks the remaining teeth and the bone that the implant options depend on.

If Mr James pushes back on cost — as he reasonably might — they do not simply accept it and move on. They restate plainly what the clinical risk of that choice is, offer what can be done within his means, and make sure the decision is genuinely informed rather than defaulted into.

Finally — and this is what separates a strong candidate — they answer the question of whether they can do this themselves as a general dentist candidly, naming the specific challenges they will face rather than presenting the plan as simple. Mr James leaves understanding why the straightforward answer he wanted is not the one that serves him best.

Replacement Options — Diabetic Patient with a Wedding in Two Weeks NEW

Prosthodontics · 11 minutes · Cluster 3

The station. *Naseem, 53, presents wanting to replace missing teeth. Missing: 26, 36, 46. Tooth 16 is slightly supra-erupted. Medical history includes diabetes, and vitamin D has been prescribed by the GP. There is a family wedding in two weeks. Take a history give short term replacement options*

A strong candidate greets Naseem warmly and opens by finding out what is driving the visit now — the wedding in two weeks — while gently establishing why the teeth have not been replaced before this point. They take a structured history rather than jumping to options: how often Naseem attends and why, why 26, 36 and 46 were removed, and what the current oral hygiene routine looks like.

On the medical history they do not simply record "diabetic". They ask how the diabetes is managed, whether recent blood tests have been done, and about the vitamin D the GP has prescribed, connecting irregular attendance to the wider pattern of care where relevant. The social history covers smoking, alcohol and — importantly — finances, since finances will decide what is genuinely achievable.

Before recommending anything, the candidate asks Naseem what he has in mind: removable or fixed. They then set out the options honestly. They identify the supra-erupted 16 and explain why it matters — a resin-bonded bridge opposing a supra-erupted tooth can be dislodged — and discuss whether to reduce the tooth prosthodontically or intrude it orthodontically, noting that diabetes already places him at higher risk and weighs against prolonged intervention.

Crucially, they explain why diabetes changes the long-term picture: reduced immunity and dry mouth make a poor environment for any prosthesis, a prosthesis is a foreign body that carries real risk of recurrent caries, and meticulous hygiene, home fluoride and more frequent reviews are what keep that risk manageable.

They use the wedding to motivate rather than to rush, saying clearly that definitive work will not be finished in two weeks while agreeing what can realistically be started. A plan along the lines of a resin-bonded bridge at 26 with a removable partial denture for 36 and 46 is reached jointly, with review arranged.

Broken Denture Teeth — Metal RPD, Reduced Vertical Dimension PQ-63

Prosthodontics · 11 minutes · Cluster 2

The station. *Mr/Mrs Cooper, 58, presents unscheduled to your practice this morning, distressed, holding a metal partial denture — several posterior teeth, including a canine, have fractured off the metal framework overnight; he discovered them on his pillow. The denture was made by a colleague about six months ago. He lost his natural teeth in an accident 30 years ago and has worn dentures ever since. Task: take a history, examine the patient, explain your diagnosis, and manage his complaint and immediate treatment needs.*

This station rewards de-escalating before diagnosing. Mr Cooper is distressed but not hostile — he softens quickly once acknowledged, and becomes more guarded and pointed ("was this made properly?") if the candidate moves straight to examination. The clinical findings — reduced vertical dimension and angular cheilitis — are not given upfront; they only surface once the candidate actually examines him or asks what was found.

OPENING "I can see you are upset — your denture teeth have broken and you've had constant trouble with this denture. How about we find out what may have caused this?" Then ask what he recalls happening, and whether this has happened with a denture before, before offering any explanation of your own: "I may have an answer, but what do you think happened here?"

HISTORY THAT MATTERS He's worn dentures for about 14 years, this is his second or third set — a pattern worth naming, since each remake has likely been built to an already-collapsed bite. Ask directly, without judgement, whether he sleeps in the denture (he does — he dislikes the feel of his gums touching) and whether his bite has felt different lately (it has — "a bit lower," food catching differently). Ask about grinding: his partner has mentioned it, he's never raised it with a dentist.

EXPLAINING THE DIAGNOSIS "You've had a history of dentures that keep breaking, and you had high hopes for what my colleague made you. This one didn't fracture completely, since it has a metal frame, but the teeth themselves are still plastic. The height of your bite has been reduced — we call that reduced vertical dimension — and on top of that, it sounds like you're grinding. I can see wrinkling and soreness at the corners of your mouth, called angular cheilitis, which happens when the bite collapses like this. When we make a denture we record how your jaws meet — if that's off, or the teeth aren't built to cope with your jaw relaxing further at night, extra force comes through and the acrylic fails where it meets the metal."

THE COLLEAGUE QUESTION Handle this diplomatically, without blame or dismissal: "My colleague may have overlooked correcting the bite, or not fully removed an interference. I'm happy to have a chat with him about this if you'd like — would you be interested in me arranging that?"

SHORT-TERM MANAGEMENT "What we can do today is fix this denture for you, so you can get back to work — but I can't guarantee a new one won't break again, especially given the grinding and night wear." Present chairside vs laboratory repair honestly: chairside is same-day and low-cost but weaker and the bite can't be checked; a lab repair is stronger and lets the bite be corrected, but costs more and takes longer. Chairside superglue is contraindicated — it doesn't bond under load, can distort the fit, and some adhesives are toxic if swallowed.

ANGULAR CHEILITIS AND NIGHT WEAR Treat with a topical antifungal (e.g. miconazole cream) and explain it will keep recurring until the vertical dimension is corrected. Recommend he stop sleeping in the denture, and offer a constructive alternative rather than just taking something away: a Michigan splint to wear at night instead.

LONG-TERM PLAN Refer to a prosthodontist for correction of the vertical dimension — an overlay denture, a remake at a verified bite height, or implant-retained options/a bridge if he's interested and it suits his budget. Tie stress management into denture-care advice, since ongoing grinding under stress is a maintainable risk factor.

LIKELY FOLLOW-UP QUESTIONS Whether the current denture can just be modified (yes short-term, but it won't fix the underlying VDO problem); options if he can't afford implants (a well-made RPD or overlay denture remains reasonable); natural teeth use canine guidance while dentures are usually set up with group function/balanced occlusion to spread load; grinding could point to sleep apnoea if there's snoring or daytime fatigue, worth asking about rather than assuming stress is the only cause; denture lifespan is normally 8–10 years, so breakage at six months is a genuine outlier that deserves a proper explanation, not dismissal.

Combination Syndrome — Mrs Cavalli PQ-64

Prosthodontics · 11 minutes · Cluster 2

The station. *Mrs Cavalli, a regular patient, presents wanting a new upper complete denture and a new lower denture. She has four remaining lower teeth (32, 31, 41, 42) with grade 3 mobility and gingival recession, which she does not want removed. She has been diagnosed with osteoporosis, and her GP has referred her to you for dental clearance before starting treatment. At a previous visit you recommended removing the four remaining teeth, but she was unwilling to proceed and has said she'd prefer to travel overseas for a second opinion and treatment. Task: explain why tooth removal is important, and address and manage her concerns.*

This station opens with the refusal already in place — she's had time to sit with the extraction advice since her last visit and has decided against it before this consultation even starts. The test is giving her genuinely new reasons to reconsider, not repeating the same advice more firmly, while handling the clearance time-pressure and the overseas idea with honesty rather than either backing down or shutting her down.

OPENING "Mrs Cavalli, I remember we spoke last time about removing your remaining four teeth, and I understand you've thought about it and would rather not go ahead. Before I say anything else, can you tell me what's been on your mind about it?" This invites her reasoning rather than immediately repeating the recommendation, and is far more likely to surface her real concern than launching straight into clinical facts again.

TWO REASONS, NOT ONE "I want to explain properly why I'm recommending this. There are really two separate reasons. First, on examination, your four remaining teeth have quite severe mobility and the gum around them has pulled back significantly — the bone supporting them has been lost to the point where they're not salvageable, regardless of anything else going on. Second, because your GP has referred you for clearance before starting your osteoporosis medication, there's a time-sensitive safety issue: I can't give clearance while these teeth are still there untreated, because if we needed to remove them after you'd started the medication, that carries a real risk of MRONJ, where the jawbone doesn't heal properly. The safe time to deal with this is now — not something we can simply monitor and revisit later."

THE MECHANISM "There's also a connection between your loose upper denture and what's happening with these four teeth. They're biting directly against your upper denture instead of against back teeth, which concentrates a lot of force right at the front of your jaw. Over time — especially with your denture being about ten years old — that's worn away the bone at the front of your upper ridge, made the gum there soft and loose, and let the front of the denture tilt, which has pushed these four teeth further out and loosened them. This is called combination syndrome, sometimes also called Kelly's syndrome. If we don't address it, your new dentures will keep failing to fit, and you may eventually need surgery to fix the bone."

A CONCRETE ALTERNATIVE "I don't want to just tell you what has to go without telling you what we'd replace it with. Once those four teeth come out, I'd really recommend an implant-retained option — either an implant-supported bridge or an implant-retained overdenture — which gives much better support and specifically stops this from happening again. It costs more, but it's the most stable long-term solution. A well-made complete denture, with the bite properly corrected and support restored at the back, is also reasonable if that suits you better."

HER UNDERLYING FEAR Ask what worries her most, rather than assuming it's the procedure itself — her deepest fear is ending up unable to eat properly and looking "sunken," having seen relatives struggle with full dentures. Naming this and addressing it with a concrete plan is the single most effective way to shift her.

THE OVERSEAS PLAN Meet it with curiosity, not dismissal: "Tell me a bit more about what you're thinking there." Once she opens up (flights looked into, a friend's experience overseas, some embarrassment about it): "I understand the appeal, especially with cost in mind, but I don't recommend having this done overseas — a course like this can't really be finished in one trip and needs ongoing review, we wouldn't have easy access to your full history here for follow-up, and neither your insurance nor my own indemnity would cover you if something went wrong. It's also worth asking whether repeated trips back for adjustments would end up costing more anyway."

DENTURE STOMATITIS AND HEALTH PROMOTION Treat the irritation under her upper denture and correct her cleaning habits: soap and water rather than toothpaste (which is abrasive and scratches the acrylic), and avoid soaking overnight in plain water (fungi thrive there).

WRAP-UP Refer to a prosthodontist for the implant discussion, and confirm the clearance pathway back to her GP once a plan is agreed. Common ways to fail this station: simply repeating the extraction advice more forcefully; giving only one reason for removal instead of both; dismissing the overseas idea outright; not

offering a concrete replacement plan; implying that starting medication and monitoring the teeth afterwards is safe.

Paediatric dentistry

Caries Risk Assessment: 8-Year-Old PQ-2

Paediatric dentistry · 11 minutes · Cluster 1

The station. *Mehmat Khan is a 8 years old cooperative child came in your practice with mother. Mother is worried about his teeth turning black. single mom busy at work, Grandparents mostly take care oh Mehmat. On examination you found a lot of carious teeth. caries on both permanent molars Consult with the mother about her concerns. Intraoral picture given for lower jaw, caries on both permanent molars, E/D/C's. Consult with the mother about caries risk assessment and treatment.*

This station is not really about caries. It is about finding out who actually feeds and supervises this child, and about being told "no fluoride" and still producing a plan.

The parent opens with "I feel terrible, I don't know what I did wrong." Answer that before anything else: "You've brought him in, and that's the thing that matters. This is really common and it's almost never one thing somebody did wrong — let's work out what's driving it and fix it together." A candidate who launches into diet questioning without acknowledging that line has already lost the room.

The parent is worried about whether this will "affect his permanent teeth." The permanent first molars are already erupted and already showing pit-and-fissure decay. That is the single most important thing to surface, gently: the damage is not a future risk, it has started. Say it without alarm, but say it clearly — the whole treatment plan depends on the parent understanding it.

The parent works and the grandparents do most of the day-to-day care. So a diet history taken from the parent alone is incomplete, and they will tell you as much if you ask. You need:

The practical point: everyone who feeds him has to be on the same plan. A diet sheet the parent follows and the grandparents do not is not a plan.

He brushes with the door shut and will not let anyone supervise. An 8-year-old does not have the manual dexterity or the attention to brush adequately unsupervised, and "does he brush?" gets you a yes that means nothing. Ask who watches, how long, what toothpaste, what time, and what happens after. Then work with his need for independence rather than against it: he brushes himself, and a parent or grandparent checks afterwards — disclosing tablets, a quick look, a chart. Frame it as him being in charge and an adult signing off, not as taking the job away from him.

The family declines fluoride on religious grounds. Do not argue, do not repeat the offer three times, and do not treat it as ignorance to be corrected. Explore what specifically the concern is, offer to provide information, and respect the decision. Then adapt:

No, and explain why rather than just refusing. Primary molars hold space; taking them out early lets the permanent teeth drift and crowd, and can create an orthodontic problem that costs far more than restoring them now. Also — and this is the point that lands — extracting baby teeth does nothing about the permanent molars, which are the teeth already decaying.

Naming the four and then populating each one is a clean, examinable way to show a complete preventive plan rather than a list of tips.

Full charting, bitewings, assessment for pulpal involvement and any pain or sepsis. Then a staged plan: stabilise the active lesions, restore, seal the permanent molars, and set a short recall (around three months

given the risk profile and the declined fluoride).

They have not been before because of cost and because a single working parent cannot easily get here. Both are solvable and neither is a character flaw. Raise the Child Dental Benefits Schedule — an eligible child gets a capped amount over two consecutive calendar years, which covers a lot of what this child needs. Offer appointment times that fit around work, and stage treatment so it is affordable. A plan the family cannot attend is not a plan.

Never asking who else feeds him. Taking "he brushes" at face value. Missing that the permanent molars are already involved. Arguing with the fluoride refusal, or ignoring it and prescribing fluoride varnish anyway. Offering GIC without noticing it releases fluoride. Agreeing to extract the primary teeth as a shortcut. Producing a preventive plan with no mention of cost or of the grandparents.

Congenital Syphilis Presenting in a Child PQ-8

Paediatric dentistry · 11 minutes · Cluster 1

The station. *Stevie Umanarah is an 8-year-old patient at your clinic, visiting today with her foster carer Mrs Kaur. The carer is mainly concerned about the appearance of her teeth. You examined her and took an intraoral picture showing notched, upper incisors. She has difficulty holding things, has partial eye sight and hearing loss. Explain the necessary investigations and address the carer's concerns.*

This station looks like a paediatric aesthetics case and is actually about consent and confidentiality. The dental findings are the easy part; the marks are in knowing what you are allowed to say and to whom.

Notched, screwdriver-shaped permanent incisors (Hutchinson's incisors) and mulberry molars. Put those alongside the partial eyesight and the hearing loss given in the brief and you have the classic triad of late congenital syphilis — Hutchinson's teeth, interstitial keratitis and sensorineural hearing loss. You should recognise it. That is not the same as announcing it.

Mrs Kaur is a SHORT-TERM foster carer, and she has not been told Stevie's underlying diagnosis. She knows only that Stevie has "some developmental issues."

It is not your place to hand her that diagnosis across the surgery. You did not make it, you have not seen the medical records, and disclosing it to someone the treating team has chosen not to inform is a breach of the child's confidentiality. Describe what you can see without naming the condition:

"What I can see is that these teeth didn't form in the usual way while they were developing — the shape and the surface are different from normal. That happens with a number of conditions that affect a child early on. I don't have Stevie's medical records in front of me, so rather than guess I'd like to contact the organisation and confirm what's already known and what's been shared with you."

That sentence — "let's confirm with the organisation" — is the one the station is looking for.

A short-term carer has authority to consent to EXAMINATION and INVESTIGATION. She does NOT have authority to consent to treatment. Treatment consent has to come from the organisation or statutory guardian, and a referral to hospital or to a specialist needs its own separate approval.

Mrs Kaur will push. There is a wedding coming up, she wants Stevie's teeth sorted, and she will offer to pay out of her own pocket. Handle it kindly and hold the line: paying for something does not create the legal authority to consent to it. Willingness to fund treatment and authority to agree to it are two different things, and only one of them is yours to accept.

"I can hear how much you want this sorted for her, and I do want to help. The difficulty isn't the money — it's that consent for treatment has to come from the organisation that has guardianship, not from us. What I can do today is examine her, take the records we need, and get that approval moving so we're ready to go as soon as

it comes through."

Examination and investigations are within her consent, so use the appointment properly:

That gives the organisation a complete picture to consent against, and means nothing is wasted.

Stevie has difficulty holding things, plus partial sight. So ask the obvious follow-up: how is she actually brushing? Then adapt — a modified or built-up handle, a powered brush, positioning, and carer assistance for the parts she cannot manage. Her caries risk is already high; a toothbrush she cannot grip is a live problem, not a footnote.

Stevie will move between carers. Every handover is a chance for her routine to be lost. So teach HER — demonstrate at her level, let her practise, give her something she owns and can carry to the next placement. Speak to her directly and at an age-appropriate level rather than talking over her to the adult for ten minutes. This is the part candidates skip, and it is explicitly in the marking.

She will raise the spacing and the puffy gums if asked. The spacing and any jaw discrepancy are complex, need specialist input, need a separate referral and separate consent, and are likely a hospital pathway. Worth knowing: orthodontic treatment is not normally covered under the Child Dental Benefits Schedule, but funding pathways do exist for children with developmental conditions — so do not tell the carer it is simply unfunded.

Naming congenital syphilis to the carer. Accepting her consent for treatment. Accepting her offer to pay as though that settles it. Never asking what kind of carer she is or what she has been told. Missing the dexterity issue. Talking only to the adult and never to Stevie. Promising treatment dates you have no authority to promise.

12-Year-Old with trauma PQ-12

Paediatric dentistry · 11 minutes · Cluster 3

The station. 12 year old Girl has injuries during hockey. Didn't wear mouth guard, 21 fracture to enamel dentin area no exposed the pulp. 11 displaced palatally showed no response to vitality test. No X-ray taken. provide short- and long-term solution.manage patients concerns.

An excellent candidate would begin by introducing themselves warmly to both Sarah and her mother, acknowledging Sarah's distress and reassuring her before proceeding. They would take a focused trauma history covering the time and mechanism of injury, whether a mouthguard was worn, presence of pain (rated 6/10, worse with cold air), any loss of consciousness, headache, nausea, or vomiting, and confirm tetanus immunisation status is current. They would note the fracture fragment was not retrieved and that bleeding from the lip has stopped.

On examination, they would perform a systematic extra-oral assessment of the face, TMJ, and lip laceration, then an intra-oral assessment including soft tissues, occlusion, mobility testing, percussion, and sensibility testing. For tooth 11, they would identify a complicated crown fracture with approximately 2mm pulp exposure, tenderness to percussion, and a vital response. For tooth 21, they would identify an uncomplicated crown fracture involving enamel and dentine without pulp exposure. Periapical and occlusal radiographs would be requested immediately to assess root development, periapical status, and rule out root fracture or alveolar injury.

The candidate would explain to Sarah and her mother in plain language that the 11 displaced palatally we call lateral luxation(not responding to cold test as the nerve could be in shock or it could be dead this need to be reviewed) . For tooth 21, they would place a glass ionomer or composite resin restoration to protect the exposed dentine. They would decline the request for antibiotics, explaining clearly that antibiotics are not indicated for dental trauma in a healthy child with no signs of infection, and that the pulp treatment itself is the appropriate management.

Post-operative instructions would include soft diet, ibuprofen and paracetamol as per body weight for analgesia, gentle oral hygiene, and an emergency contact number. A follow-up schedule would be arranged at 2 weeks to remove splint, 1 month 2 months 3,6 12months yearly for upto 5 years, monitoring for pulp necrosis, root resorption, and periapical pathology.

Mouthguard fabrication would be strongly recommended before Julie returns to sport.

Five-Year-Old with a Draining Sinus PQ-15

Paediatric dentistry · 11 minutes · Cluster 1

The station. Child 5 years old girl has mild discomfort on 64/65 which was filled with composite/GIC by school dental van nurse few months ago. Mum also noticed sinus above the tooth recently. Gather information and Answer parent's concern, no need to give treatment plan or diagnosis.

An excellent candidate begins by warmly introducing themselves to both Sarah and Sally. They establish rapport with Sally using age-appropriate language — perhaps commenting on something she is wearing or holding — before directing most clinical questions to Sarah while keeping Sally engaged. The candidate takes a focused history: asking when the 'pimple' first appeared, how long it has been present, whether it has ever discharged, and whether Sally has had any associated pain, fever, facial swelling, or difficulty eating or sleeping. They elicit the history of a toothache approximately four months ago that resolved spontaneously — a critical detail suggesting pulp necrosis. The candidate asks about medical history, allergies, current medications, and immunisation status, confirming Sally is fit and healthy with no contraindications to dental treatment. They then address Sarah's concern about diet and caries risk in a non-judgmental way, explaining that early childhood caries is multifactorial — frequency of fermentable carbohydrate exposure, oral hygiene habits, saliva, and tooth morphology all contribute — and that cordial at kindergarten, even if perceived as low-sugar, can be a significant risk factor. The candidate explains that the 'pimple' is likely a sinus tract, a small channel the body has created to drain infection from the tip of the tooth root, and that this indicates the nerve inside the tooth has been affected. They clarify in plain language that a draining sinus actually means the infection is currently releasing pressure, which is why Sally is not in acute pain, but that this does not mean the infection has resolved — it requires definitive dental treatment. The candidate reassures Sarah that antibiotics alone would not fix the problem, as the source of infection remains inside the tooth, and that a clinical examination is needed to determine the best course of action. They gain consent for examination using calm, positive language with Sally. walk through the management today.

Uncooperative Four-Year-Old Child: Charting and Diet Counselling PQ-29

Paediatric dentistry · 11 minutes · Cluster 2

The station. Mrs McConnell is visiting your clinic today for her 4year old Paula complaining about pain in 1 tooth on lower left side Bitewing radiographs (attached, left and right — taken despite initially uncooperative behaviour): - Right bitewing: proximal caries into dentine, mesial surface of 84; occlusal caries into dentine, 85. - Left bitewing: proximal caries into dentine, distal surface of 64; early enamel-only proximal caries, mesial of 75. - No periapical pathology or pulpal involvement evident on either film. Diet chart (attached, one typical day as recorded by Mrs Chen): - 7:00am: cereal with milk - 10:00am (daycare): dried fruit and a muesli bar - 12:00pm (daycare): sandwich, apple juice box (#2) - 3:00pm (daycare): cereals with honey - 6:00pm: dinner baked beans - 7:00pm: apple juice box (#4) - 7:30pm: brushing (once daily, fluoride-free "natural" toothpaste) - 8:00pm: Sultana after dinner - Water intake: occasional tap water only Give your diagnosis and manage mothers concerns. Consedering Shes uncooperative discuss about behaviour management give treatment plan as the patient lives 200km away and cant visit frequently.

An excellent candidate would begin by warmly introducing themselves to both the mother and Paula, using simple, friendly language to reduce anxiety.

identifies the risk factors from the history and details provided and explains the diagnosis (dental caries and High risk of dental caries) and pathophysiology of the disease explains the impact of the risk factors in her case and then working towards the management.

Once Paula allows a brief visual examination, and correlating bitewing findings showing caries in primary molars. review diet chart and then explain these risks clearly

Management

if successful in behaviour management (ask parent for suggestions/ tell show do/ cool names/ Distraction/ modelling) - Xray 1 temporary filling GIC/ZOE or Hall crown - Xray 2 RCT specialist or extraction and space maintainer

if unsuccessful - Refer to specialist (inhalation sedation or tablet) - Try one more time next time if still not cooperative ref to hospital to be treated under GA explain do nothing option as leaving it untreated can cause infection severe pain and swelling

health promotion 3monthly reviews as hes high risk working as a team to improve his oral health On oral hygiene, the candidate would address the fluoride toothpaste concern directly, reassuring the mother that a low-fluoride toothpaste (400–550 ppm, such as Colgate Minifluoride) (if anti fluoride due to religious belief respect that and suggest GC tooth mousse) is safe and recommended for children under six, with a pea-sized amount and spitting encouraged. Brushing should occur twice daily, with the parent brushing or supervising until age seven to eight. The candidate would arrange a review appointment via tele dentistry as the patient lives far explain the importance of regular dental visits for monitoring and preventive care, and close with empathy and encouragement.

3-Year-Old with Trauma PQ-34

Paediatric dentistry · 11 minutes · Cluster 2

The station. Max Smith 3-year-old boy is here with his father today complaining of tooth that has become grey. fell down a week ago from the swing while and had a small cut on the lip, mom applied pressure and bleeding stopped. Yesterday he noticed the black discoloration in front tooth (61). Explain diagnosis, treatment plan and address parents questions.

An excellent candidate would begin by warmly greeting Max's father, introducing themselves and establishing rapport with both him and Max. They would take a structured trauma history: confirming the injury occurred a week ago, and how it happened. They would screen for head injury red flags — asking specifically about any loss of consciousness, vomiting, or nausea following the fall — and would confirm his tetanus immunisation status is up to date, noting that the lip laceration was managed at the time with simple pressure and has since resolved. For the examination, they would use a tell-show-do approach to keep Max calm and would involve his father throughout. They would examine tooth 61 — the upper left primary central incisor — and identify the grey discolouration as consistent with pulp necrosis following an intrusion injury sustained a week earlier, and would also note tooth 51 has been intruded, checking the occlusion, mobility, and surrounding soft tissues of both teeth and taking an appropriate radiograph to assess the extent of the intrusion. They would explain the diagnosis to the father in plain, reassuring language: that tooth 61 has been affected by dental concussion and now looks grey, while tooth 51 has been pushed up into the gum by the fall, and that the dark colour of tooth 61 tells us the nerve inside has been affected. Because these are baby teeth, the candidate would explain that active monitoring rather than immediate treatment is the right approach — tooth 51 may reposition itself spontaneously over the following months, and any greyiness in tooth 61 needs to be watched rather than treated straight away. They would explain the possible outcomes clearly: for the intruded tooth, spontaneous repositioning is the hoped-for course, but signs of infection such as pain, a gum boil, or increasing mobility would need review, though damage to the underlying permanent successor is possible though less likely at this age; for the grey tooth, the options going forward — should they become necessary — range from continued monitoring, to a tooth-coloured filling, to root canal treatment (preferably by a specialist) or extraction with a

space maintainer if the nerve becomes infected. They would advise a soft diet, age-appropriate analgesia such as paracetamol or ibuprofen, and gentle oral hygiene with a soft toothbrush. They would counsel the father on potential longer-term sequelae — further discolouration, infection or abscess, and importantly the possibility of damage to the developing permanent successor tooth, including hypoplasia or dilaceration — and would explain that signs of infection such as swelling, a sinus tract, pain, or failure of the tooth to re-erupt would be the trigger for extraction. They would arrange a structured follow-up schedule at one week, six to eight weeks, six months, and yearly until the permanent tooth erupts, with clear safety-net advice to return urgently if swelling, pus, or systemic illness develops, and would reinforce the general importance of supervising young children around play equipment. They would offer written, scheduled follow-up reminders, and if Max were not present at a scheduled review, would emphasise booking the next appointment as early as possible rather than letting it lapse.

MIH 8-Year-Old — Sensitivity & Appearance PQ-53

Paediatric dentistry · 11 minutes · Cluster 2

The station. 8-year-old child, attending with mother. Chief Complaint: Mother reports sensitivity in tooth 26 (upper left first permanent molar). She is also concerned about the appearance of the child's front teeth. Examination Findings / Clinical Images: Image 1: Tooth 26 with a well-demarcated yellow-brown opacity and post-eruptive enamel breakdown on the occlusal surface. Image 2: Upper permanent incisors showing creamy-white to yellow opacities with sharp borders against adjacent normal enamel, distributed asymmetrically — 11 and 21 affected, 12 and 22 largely spared. The remaining erupted permanent teeth are unaffected and the primary dentition is sound. Task: Explain the diagnosis to the mother, including the differential diagnoses you have considered and excluded. Address both of her concerns — the sensitivity in tooth 26 and the appearance of the front teeth — and discuss management options for each.

An excellent candidate begins by warmly greeting both Jenny and Sophie, addressing Sophie directly with an age-appropriate comment to establish rapport and signal that Sophie is the patient. The candidate then takes a structured history, asking Jenny about Sophie's early childhood health — specifically recurrent illnesses and antibiotic use in the first three years of life — eliciting the history of four to five ear infections treated with amoxicillin. The candidate also asks about pregnancy history, eliciting the second-trimester antibiotic course, and importantly does not over-attribute causality to it, recognising that this exposure falls outside the mineralisation window for the affected teeth. Crucially, the candidate asks about family history of similar-looking teeth in parents or siblings — eliciting a clear negative — and takes a fluoride history, establishing optimally fluoridated water, age-appropriate toothpaste and no supplements.

The candidate then explains a single diagnosis: Molar Incisor Hypomineralisation, affecting tooth 26 and the upper central incisors. They explain in accessible language that the enamel on these particular teeth did not form fully while it was developing in Sophie's first few years, that this is linked to the period of recurrent infections, and that only these teeth are affected because only these teeth were forming at that time. The candidate demonstrates diagnostic reasoning by explicitly considering and excluding the two principal differentials: Amelogenesis Imperfecta is ruled out because it is hereditary and would be expected to affect all the teeth in both the baby and adult sets and to appear in other family members, whereas Sophie's baby teeth and most of her adult teeth are sound and the family history is negative; dental fluorosis is ruled out because it produces diffuse, symmetrical mottling across all teeth developing at the same time rather than the sharply demarcated, asymmetric patches seen here, and because the fluoride history is unremarkable. These exclusions are explained to Jenny in lay terms rather than recited as a list.

The candidate explicitly addresses Jenny's guilt, explaining that this condition is not caused by anything she did or did not do during pregnancy or afterwards, and that the childhood ear infections were beyond anyone's control. The emotional concern is validated before reassurance is offered.

For tooth 26, the candidate recommends fluoride varnish, a protective restoration using glass ionomer or composite resin given the post-eruptive breakdown and sensitivity, desensitising advice, and three-monthly

review given elevated caries risk, noting that MIH-affected molars can be harder to anaesthetise and need long-term monitoring. For the incisors, the candidate explains that definitive cosmetic treatment such as microabrasion, resin infiltration or composite veneers is deferred until Sophie is around sixteen to eighteen years old when facial growth is complete and the gum margins are stable, with interim fluoride varnish and avoidance of abrasive toothpastes, and offers genuine reassurance that the appearance can be properly addressed in time. Sophie is addressed directly at least once during the management discussion.

MIH in an 11-Year-Old — Second Opinion on GA Extraction PQ-59

Paediatric dentistry · 11 minutes · Cluster 2

The station. 11-year-old boy, attending with his parents. Chief Complaint: Sensitivity to cold in his back teeth. History: Mother took him to a public hospital previously. General dentist at the hospital suggested extracting the back teeth under general anaesthesia. Parents seeking a second opinion. Examination Findings / Clinical Image: Photograph showing hypomineralised (chalky/opaque) first permanent molars and incisors. Baby teeth appear normal. Diagnosis: Molar Incisor Hypomineralisation (MIH) — affects permanent molars and incisors, primary dentition unaffected. Task: Diagnose and manage. Address parent's concern about the hospital dentist's recommendation for GA extraction.

An excellent candidate begins by warmly greeting Liam directly — not just his parents — and explains the purpose of the visit in simple terms. They take a focused history, asking Liam whether the sensitivity is triggered by cold or occurs on its own, and elicit that there is no spontaneous or nocturnal pain. They ask Wei about early childhood illnesses and acknowledge the hospitalisation for bronchiolitis at eight months as a plausible aetiological event during the critical window of first permanent molar mineralisation. They reassure both parents clearly and empathetically that MIH is a developmental condition — not caused by diet, sugar, brushing habits, or anything done during pregnancy — and that this reassurance is delivered as a genuine emotional acknowledgement, not a throwaway line. The candidate names the diagnosis as Molar Incisor Hypomineralisation, explains in plain language that the enamel did not harden properly during early childhood, distinguishes it from decay (no bacterial cause), fluorosis (no excess fluoride), and amelogenesis imperfecta (not hereditary, primary teeth unaffected — as seen in the photograph). They present a staged conservative plan: high-fluoride toothpaste and desensitising agents first, fissure sealing of at-risk surfaces, GIC or composite restorations for any areas of surface breakdown, and Hall-technique preformed metal crowns for significantly broken-down molars. They explain that extraction under GA is reserved for teeth with irreversible pulpal involvement or that are unrestorable — neither of which applies to Liam today — and frame the hospital dentist's recommendation diplomatically, acknowledging resource constraints without criticism. They address David's questions about cost and prognosis honestly. They set a three-to-six-month recall and explain that MIH teeth can deteriorate, so monitoring is essential. Throughout, they speak to Liam in age-appropriate terms and acknowledge his embarrassment about his front teeth.

Caries Risk Assessment in a Cooperative 8-Year-Old PQ-60

Paediatric dentistry · 11 minutes · Cluster 2

The station. Mehmet Askoff, an 8-year-old boy, attends today with his father. His father is worried about black holes in Mehmet's teeth. On examination you find multiple grossly carious primary molars — an intraoral photo is provided, and a bitewing radiograph may sometimes be given. Mehmet is a cooperative child. Conduct a consultation with the father covering the caries risk assessment and management.

****Opening and rapport**** Greet Mehmet directly first ("Hi Mehmet, I'm just going to have a look at your teeth in a minute — you're going to do a great job") before turning to his father for the history — this signals to the

examiner that the child is included, not just discussed over his head. Acknowledge the father's worry without minimising it: "Thanks for bringing him in — let's work out together what's going on and what we can do about it."

****Structured history**** Take a focused caries-risk history rather than a generic diet question:

In this case: frequent juice/cordial, afternoon snacking on lollies/chips, once-daily unsupervised brushing, non-fluoridated bore water, and an 18-month gap since the last dental visit — a cluster of modifiable risk factors, not one single cause.

****Explaining the diagnosis**** Use the photo (and bitewing, if taken) to show the father what's happening: "These dark areas are decay — the bacteria in the mouth are feeding on the sugars from food and drink and producing acid, which breaks down the tooth." Make the mechanism concrete and non-judgemental: it's the **frequency** of sugar/acid exposure through the day that overwhelms the mouth's ability to repair itself between attacks, not simply "too much sugar" in a moral sense. Explicitly reassure the father this is common and very fixable, not a reflection of bad parenting.

****Answering "why did this happen when I thought we were doing okay?"**** Name the specific contributing factors identified in the history (frequent sugary drinks, afternoon snacking, once-daily brushing, non-fluoridated water, the gap since the last check-up) as a combination, and frame the consult as identifying what to adjust rather than assigning blame.

****Answering "can we just watch them since they're baby teeth?"**** This is the key inflection point of the station. Explain clearly why untreated cavitated primary molars still need active management: they can progress to pulpal involvement causing pain and abscess, a dental abscess in a young child is a genuine medical event (swelling, missed school, possible antibiotics or emergency extraction), premature loss of a primary molar can cause space loss and crowding for the permanent successor, and ongoing pain affects eating, nutrition, and sleep. "Baby teeth hold the space and the bite for years yet — some of these won't be replaced until he's 10 or 11 — so treating decay in them now protects both his comfort today and his adult teeth later."

****Management plan****

****Answering "how do we protect his other/adult teeth?"**** Tie the prevention plan directly back to this question: reducing exposure frequency, fluoride at the right strength and frequency, sealants on permanent molars as they come through, and closer recall until the risk factors are under control.

****Closing**** Summarise the plan in plain terms, check the father's understanding, and give him one or two concrete, realistic first steps to start with (e.g. "swap the after-school snack for something like cheese or fruit, and keep sugary drinks to mealtimes only") rather than an overwhelming list all at once.

Orthodontics

Missing Permanent Lateral Incisor PQ-21

Orthodontics · 11 minutes · Cluster 1

The station. 7-year-old Olivia Smith attends with her mother. Her primary (baby) lateral incisor fell out about a year ago and the permanent lateral has not erupted yet. Her parents are concerned about the gap in the middle and the space between the central incisor and canine. An intraoral photo shows a large midline diastema. photo shows a deep bite Find out more about her situation and answer the parents' queries — explain the reasons for this and the necessary investigations.

An excellent candidate would begin by warmly greeting Mrs Smith and Olivia establishing rapport before taking a focused history. They would confirm the timeline of the primary lateral incisor exfoliation (approximately one year ago), ask whether any other baby or adult teeth have been slow to erupt or missing, and specifically

enquire about a family history of missing or small teeth — noting that hypodontia has a strong genetic component. They would ask about olivia's general health, her dental history.

On examination, the candidate would assess olivias facial profile and skeletal relationship (Class 2), lip competence, and arch form. They would note the spacing in the upper arch, assess the mobility of the retained upper left primary lateral incisor (62), and examine the occlusion.

The candidate would explain the diagnosis and reasons for diastema to Mrs Smith in plain language

possibilities for the lateral incisor at this age

explains the investigations to be carried out extraoral intraoral - palpate / check for bulge/ frenum attachment/ perio xray photos study scans or models

since they are worried about xray causing organ damage explaining that normally we recommend an occlusal xray in this situation but since they have concerns about her organ an occlusal xray is directed towards her organ we can manage with 2 IOPAs but that will require 2 xray instead of 1 (autonomy)

explains that the spaces of 1 to 2 mm is normally there at this age anything more than that is less likely to close on its own requiring referral. The candidate would acknowledge the mother's concerns about cost and appearance empathetically and outline a clear referral plan.

explains when asked what will the specialist do? early interceptive orthodontics and opg if we find those teeth are missing on small xray.

ends with review and health promotion supervised brushing and flossing) referral letter to orthodontist. future ortho treatment will put her at higher risk of decay

Assessment and Management of a Missing Maxillary Canine PQ-44

Orthodontics · 11 minutes · Cluster 1

The station. Julie, 10-year-old girl, attending with her mother. Chief Complaint / Presentation: Bilaterally missing canines in both jaws. Mum is worried about the child's appearance because of splayed front teeth. The front teeth are described as 'too big.' Eye teeth are missing from both jaws. Parent can feel a bulge in the lower jaw assuming teeth are erupting. There is no bulge in the upper jaw. Clinical Image: Intraoral photo of a child's dentition showing marked spacing/splaying of the front teeth, with missing canines, and prominent gingival/bony bulge in the lower jaw. Task: Gather information and explain what to do.

An excellent candidate would begin by warmly greeting Julie (Chloe) and her mother Sarah, confirming the child's age and establishing rapport. They would take a focused history covering the timeline of tooth eruption — specifically when the upper left canine erupted, whether any trauma has occurred, any pain or swelling, and family history of similar problems. They would ask about habits such as thumb-sucking and confirm medical history, medications, and allergies.

On examination, the candidate would describe palpating both the buccal and palatal sulci in the upper arch to localise the impacted canine, assess the mobility of the retained deciduous canine (53), evaluate arch space and crowding, and check the occlusion. They would note the absence of a buccal bulge in the upper jaw — a key clinical sign suggesting palatal impaction.

The candidate would request an OPG and correctly interpret it: identifying the upper right permanent canine (13) as palatally impacted, lying horizontally at the level of the apices of 12 and 14 with the crown pointing mesially, and confirming no root resorption of adjacent teeth and no associated pathology. They would note that 23 has already erupted normally.

Explaining findings to Sarah in plain language, the candidate would say the adult eye tooth is stuck behind the roof of the mouth and will not come through on its own. They would explain that early treatment — ideally before age 13 — gives the best chance of success. Treatment involves removing the baby tooth, referring to an orthodontist and oral surgeon for surgical exposure and bonding of a small bracket to guide the tooth into position over approximately 18–24 months. Risks of no treatment include damage to the roots of neighbouring teeth, cyst formation, and ongoing spacing concerns. The candidate would acknowledge Sarah's concerns about cost and appearance, note that private health insurance with orthodontic cover may offset costs, and arrange prompt referral.

Oral medicine

Ulcer on lateral border of tongue PQ-20

Oral medicine · 11 minutes · Cluster 1

The station. Mrs Cooper 52 year old school teacher, attending today complaining of Ulcer on right side of tongue, present for 2 months her last dental visit was 2 years back. She is fit and healthy. you havent done any examination yet. gather information and Explain the differential diagnosis, manage patient's concern, discuss further investigations needed.

An excellent candidate would begin by warmly greeting Mrs Cooper and establishing rapport before taking a focused history. They would clarify the duration of the ulcer (5 weeks), and ask specifically about pain, bleeding, and any change in sensation. They would systematically screen for red-flag symptoms including dysphagia, voice change, unexplained weight loss, and night sweats.

In a non-judgmental manner, they would elicit a detailed tobacco history (15–20 cigarettes daily for 45 years, reduced recently due to cough) and alcohol history (2 standard drinks most evenings, exceeding Australian guidelines), acknowledging these as important contextual factors without shaming the patient. They would note the absence of previous oral ulcers and the family history of lung cancer in a sibling.

examination, they would describe a 1 cm × 1 cm ulcer on the right lateral border of the tongue (middle third) with local trauma source. They would note the patient's comment that his denture feels tight complete examination explained including extraoral / intraoral / picture / xray of the tooth with dislodged filling and its management as per xray. referrals OMDR biopsy/ GP referral for chest Xray to assess cough and blood tests.

Bilateral cervical lymph node palpation would. The candidate would recognise that the combination of duration exceeding three weeks, lateral border of tongue and cough present. They would communicate this concern to Mrs Cooper with empathy and clarity, avoiding false reassurance, and explain that an urgent referral for incisional biopsy is required — ideally within two weeks in accordance with Australian Head and Neck Cancer guidelines. They would not adopt a watch-and-wait approach.

Assessment and Management of Palatal Swelling

PQ-26

Oral medicine · 11 minutes · Cluster 1

The station. Mr Michael Jones comes for check-up for painful tooth, he is new to your practice. He is seeing you today and his last check-up was 4 years ago. Large Bluish/dark red swelling on the right side of the palate between 3-7 for 5 years. Complaints about his previous dentist who did RCT filling 4 year ago. How would you gain more information and what special investigation would you do and based on the findings? Give differential diagnosis.

An excellent candidate would begin by warmly introducing themselves and establishing rapport with Mr Jones, acknowledging his presenting concern first. They would take a structured history of the presenting complaint, and when mentioned take history of the swelling has been present for five years on the right palate, asking about onset, rate of growth, any change in size, associated pain, bleeding, discharge, or functional symptoms such as difficulty swallowing, speech changes, or nasal symptoms.

They would explore the patient's concern about his previous dentist and the root canal treatment performed four years ago, clarifying which tooth was treated and whether any symptoms followed.

explore if the swelling is constant or comes and goes or reduces in size (helps to decide the differential diagnosis)

Relevant medical history would include smoking status, alcohol use, any previous malignancy, current medications, and allergies. Systemic symptoms including fever, weight loss, and night sweats would be specifically excluded. On examination, the candidate would systematically palpate the swelling, assessing size, consistency (firm), surface texture (smooth), tenderness, and fluctuance.

They would examine adjacent teeth 25, 26, and 27 for vitality using an electric pulp tester or cold test, assess for tenderness to percussion, and inspect for any sinus tract. Cervical lymph nodes would be palpated bilaterally.

Based on findings — the candidate would formulate a differential diagnosis led by dental abscess (as the swelling comes and goes) a minor salivary gland neoplasm, pleomorphic adenoma, with malignant salivary gland tumour, and Mucocele (sinus tumor, Hemangioma (red colour) lymphoma (bluish colour) .

management 1) if dental cause is found we can treat it. 2) if it is cyst or tumor referral to Maxillofacial surgeon 3) Mucocele or Hemangioma referral to oral medicine specialist

The candidate would sensitively address the patient's cancer concerns, provide safety-netting advice regarding rapid growth, pain, or dysphagia, and document findings thoroughly.

health promotion about uncontrolled Blood pressure importance of regular review.

Ulcer on the gums PQ-33

Oral medicine · 11 minutes · Cluster 1

The station. Mr. Abbott is a 65-year-old patient, attending your surgery for the first time today asking for a lower denture to replace his missing teeth. You examined the patient, and you find that the patient is wearing an upper complete denture for the past 8 years with no lower denture. You notice angular cheilitis on the corners and You also notice an ulcer; the patient says that the ulcer has been present there for the last 2 months since he had his lower canine removed. The ulcer is about 2-3mm in size, it's not bleeding, Xray shows a mass in that area and patient does not experience any pain. He is leaving for work after 3 months. He won't be back until 3 months. obtain details and manage the patients concerns.

An excellent candidate would begin by introducing themselves and establishing rapport with Mr Abbott, acknowledging his concern about the new denture and exploring his expectations of it. They would then take a focused history covering the duration of the ulcer (two months), the precipitating trauma of the lower canine extraction, its current pain characteristics (non-tender by his report), and whether he has experienced similar ulcers before, explaining to him why each of these questions is relevant as they go, to keep him engaged rather than simply interrogated. They would specifically screen for red-flag symptoms — difficulty swallowing, unexplained weight loss, tingling or numbness, and any other oral lesions. They would explore his social history, including smoking, alcohol use, and diet, and would note his occupation as a miner as relevant to a potentially stressful lifestyle, alongside asking about systemic symptoms such as fever and his general health, including how recently he has seen his GP. The candidate would present a differential diagnosis covering a traumatic ulcer (from a denture clasp or wire, or an ill-fitting denture rubbing the area), a non-healing extraction

socket (given the extraction on the same side), and a chronic or more serious cause that needs to be considered given the two-month duration, explaining the relevance of each risk factor as they move toward what needs to happen today. They would explain the investigations needed: an extraoral examination of the face, neck, jaw joints and lymph nodes; an intraoral examination of the ulcer itself — its site, size, shape, base, borders, and the surrounding tissue and denture-bearing surface; clinical photographs for the record; and an OPG. They would explain the management plan, including referral to a specialist for biopsy and to his GP for review and blood tests, and would mention that the Australian system allows fast-tracked specialist appointments — typically within fourteen days — when cancer is suspected, so that Mr Abbott understands the referral reflects appropriate caution rather than alarm. Given his upcoming three-month absence for work, the candidate would emphasise the importance of attending this review before he leaves rather than waiting until he returns. They would provide health promotion advice, gently exploring how he has been coping generally and mentioning support resources such as 'it's ok to talk' if relevant, and would give clear safety-netting advice about what to watch for and when to seek care sooner while he is away.

Management of Recurrent Aphthous Ulceration PQ-43

Oral medicine · 11 minutes · Cluster 1

The station. *Charlotte, 20-year-old student. Chief Complaint: Ulceration for the past 3 days. She is a law student in a lot of stress because of her final exam starting in 2 months. Her sister gets similar ulcers as well. Examination Findings: Ulcer found on the inner side of her lips. Clinical Image: Intraoral photo showing an ulcer on the inner mucosal surface of the lip — consistent with aphthous ulceration. Task: Explain what special test you will do and give your differential diagnosis along with your provisional.*

I would begin by introducing myself to Charlotte and establishing rapport, acknowledging the discomfort she is experiencing. I would take a focused history exploring the frequency, duration, site, size, and number of ulcers, noting she reports episodes every 8–12 weeks lasting 7–10 days, with the current ulcer present for 3 days. I would ask about potential triggers including stress, dietary factors, trauma, new toothpaste or oral care products, and whether episodes correlate with her menstrual cycle. I would enquire about family history of similar ulcers, which is relevant given her sister is affected. To exclude systemic associations, I would ask about gastrointestinal symptoms such as diarrhoea, bloating, or weight loss to screen for coeliac disease or inflammatory bowel disease, as well as genital ulcers, ocular symptoms such as uveitis, and skin lesions to exclude Behçet's syndrome. I would confirm she is not on any regular medications and has no known medical conditions. On examination, I would describe the ulcer as approximately 5 mm in diameter, round, shallow, with a yellow-grey fibrinous base and erythematous halo on the labial mucosa, with no other lesions present. My provisional diagnosis is recurrent minor aphthous ulceration, a benign condition likely triggered by stress and possibly a genetic predisposition given the family history. For special investigations, I would request blood tests including full blood examination, iron studies, serum B12, folate, and coeliac serology to exclude haematonic deficiencies and coeliac disease as underlying contributors. Management would include topical corticosteroids such as triamcinolone acetonide paste, benzydamine mouthwash for symptomatic pain relief, and a protective barrier gel. I would advise stress management strategies, avoidance of known trigger foods, and good oral hygiene. I would reassure Charlotte that this is a benign condition, but advise her to return if any ulcer persists beyond three weeks, increases in size, or if systemic symptoms develop, as further investigation or specialist referral would then be warranted.

Incidental White Lesion — Differential Diagnosis & Biopsy Decision PQ-50

Oral medicine · 11 minutes · Cluster 1

The station. Hanna, 45-year-old, new patient. Attending for the first time requesting a routine check-up and clean. *Incidental Finding:* On intraoral examination, you notice a white patch on the right side of the mouth (buccal mucosa / lateral tongue region adjacent to an amalgam restoration on the right side). Patient is not aware of it. History: Smokes occasionally. Currently under a lot of stress. No other significant medical history mentioned. *Clinical Image:* Intraoral photo showing a white lesion on the buccal mucosa/lateral border — requires differential diagnosis. *Task:* Gather information. Determine what special investigations are required. Explain findings and provide a differential diagnosis.

An excellent candidate begins by systematically describing the incidental white lesion before initiating history-taking. They note the site (right buccal mucosa adjacent to the amalgam restoration), approximate size, colour (homogeneous white), surface texture (slightly raised or flat), border definition, and critically confirm that the lesion cannot be wiped off with gauze — distinguishing it from pseudomembranous candidiasis. The candidate then takes a targeted history, sensitively informing Hanna of the finding without alarming her. They elicit her smoking history accurately — not accepting 'social smoker' at face value but quantifying it as five to ten cigarettes per day over more than twenty years — and ask about alcohol consumption, revealing approximately six to eight standard drinks per week. They ask specifically about previous oral changes or white patches, uncovering the prior episode on the left buccal mucosa two years ago that resolved spontaneously — a critical red flag for field cancerisation. They confirm the absence of pain, bleeding, dysphagia, or difficulty eating, and note the proximity of the lesion to the amalgam restoration as a potential aetiological factor. The differential diagnosis is presented in a clinically reasoned order: lichenoid reaction to the adjacent amalgam is the most anatomically plausible; frictional keratosis from cheek biting under stress is reasonable; leukoplakia is the diagnosis of exclusion; and oral squamous cell carcinoma must be named as a condition to exclude, framed carefully and without catastrophising. The management plan includes immediate clinical photography and measurement for baseline documentation, consideration of toluidine blue staining, and urgent referral to an oral medicine specialist or oral and maxillofacial surgeon for biopsy within two weeks. Amalgam replacement is discussed as a supplementary diagnostic manoeuvre, not a substitute for referral. The candidate communicates honestly and compassionately, acknowledging Hanna's anxiety, explaining that most white patches are benign, and reassuring her that prompt investigation is the right and responsible course of action.

Enamel Defects in a Young Adult — Wants Veneers

NEW

Oral medicine · 11 minutes · Cluster 2

The station. Ms Tara Beckett, 24, a graphic designer, has come to your practice unhappy with the appearance of her upper front teeth. She says they have "always been patchy and pitted" and she wants veneers before her sister's wedding in three months. On examination the permanent incisors and first molars show creamy-yellow to brown opacities with horizontal grooving and pitting; the pattern is symmetrical across all four quadrants. There is no caries and her oral hygiene is good. Take a history, explain what you think is going on, and discuss management.

This station looks cosmetic and is not. Tara has undiagnosed coeliac disease, and her enamel is the visible record of it. The whole station turns on whether the candidate asks why the enamel looks like this before agreeing to cover it with porcelain.

She came in about her smile and a family wedding. If the first thing you do is redirect to her bowels, you lose her. Something like: "I can see why you're bothered by them, and I do think we can improve how they look. Before we talk about how, can I ask a bit about the teeth themselves and about your general health? The way they've formed tells us something, and it'll change what I'd recommend."

The finding is not "staining" — it is a pattern. Creamy-yellow to brown opacities with horizontal grooving and pitting, affecting the permanent incisors and first molars, SYMMETRICAL across all four quadrants and matching left to right. Canines and premolars are largely spared.

That symmetry is the diagnostic feature, and it is what separates this from the usual suspects:

Symmetrical and chronological means the insult was systemic and time-limited — it hit whatever was mineralising at the time, in every quadrant at once. In a young adult, that points at early childhood illness or malabsorption, and coeliac disease belongs high on that list.

None of this is volunteered. Ask specifically:

Your own Therapeutic Guidelines corpus makes the ulcer link explicitly: coeliac disease is listed as one of the conditions to investigate in recurrent aphthous ulceration, alongside iron/B12/folate/zinc deficiency, ulcerative colitis, Behçet syndrome and PFAPA (TG Oral and Dental, p161).

Plain language, no jargon: "Your teeth formed like this when you were small — the enamel was being laid down and something interfered with it. On its own that could be a few things. But put together with the ulcers you've always had, being anaemic twice and the iron not really fixing it, the tummy symptoms, and your sister having coeliac — I think it's worth getting you tested for coeliac disease. Your teeth may have been the first sign of it."

GP, for coeliac serology — tissue transglutaminase IgA, with a total IgA (because selective IgA deficiency gives a false negative). Confirmation is medical, usually a duodenal biopsy in adults, and is the GP's or gastroenterologist's call. Spotting it and routing it is the dentist's job; diagnosing it is not.

She will tell you she is planning to go gluten free next week because a friend felt better on it. Stop that:

"Please don't start yet — and this is the one thing I really want you to take away. The test for coeliac disease only works while you're still eating gluten. If you cut it out first, the blood test and the biopsy can come back normal even if you do have it, and then you're stuck not knowing. Keep eating normally until your GP has tested you. After that, if it's positive, gluten free for life — but let's get the answer first."

This is the trap the station is built around, because the intuitive kind response — "sure, give it a go, can't hurt" — is the harmful one.

"Going gluten free will help your gut, your energy and your ulcers. It won't change these teeth. The enamel formed the way it formed and it doesn't repair itself — improving how they look is a dental job, not a dietary one." Teeth that mineralise after diagnosis can form normally; hers are already through.

Don't commit to veneers today, and say why — not to withhold treatment, but because sequencing matters and you don't start irreversible work on someone mid-diagnosis. Then be concrete about the options so she doesn't hear a flat no:

Three months is workable for definitive treatment. Say so, but don't over-promise a result that depends on how she responds to the earlier steps.

Discussing veneer shades for ten minutes and never asking why the enamel is defective. Calling it "staining". Listing only fluorosis and MIH and never reaching for a systemic cause. Never asking about family history. Agreeing that gluten free sounds like a good idea, or letting it pass unchallenged. Telling her a gluten-free diet will improve her teeth. Refusing treatment outright without offering a pathway, so she leaves feeling dismissed about the thing she actually came in for.

Medical emergency

Uncontrolled Diabetic Patient with cellulitis PQ-28

Periodontics · 11 minutes · Cluster 2

The station. Mr Hiess is a 47 year old new patient in your clinic, complaining of severe pain and facial swelling related to his upper right side for past 3/4 days. He went to see GP before; he gave him antibiotics but it didn't seem to work so GP referred him to see you. Examination reveals that the upper right first molar is badly carious and beyond restoration, and now it is indicated for removal.. Swelling is still there, and he is in a lot of pain. His medical record shows that he has diabetes. How would you plan his management and gain consent for his treatment plan and discuss possible complications that may happen?

An excellent candidate would begin by introducing themselves and establishing rapport with Mr Hiess, acknowledging that he is in significant pain and that this must be a distressing few days. They would take a focused history covering the onset and progression of his facial swelling, and would specifically screen for systemic red flags — fever, malaise, difficulty swallowing, or shortness of breath — any of which would indicate airway compromise requiring urgent ambulance transfer to hospital; in his case, the swelling is confirmed to be stable and not threatening his airway. They would elicit a detailed diabetes history, including his current medications, adherence, his most recent HbA1c, his last blood glucose reading, and the time of his last meal and drink, recognising that all of these directly affect anaesthetic risk, surgical risk, and prescribing decisions. Intraorally, they would identify the grossly carious upper right first molar as the likely odontogenic source and assess for vestibular swelling or discharge of pus. Given the combination of spreading facial cellulitis, a fever of 38.4°C, diabetes of uncertain control, and failure to improve on oral antibiotics from his GP, the candidate would recognise this picture as one of probable immune compromise and would arrange to confirm his current HbA1c status with his GP before finalising the treatment plan. They would explain that management depends on that result: if his diabetes turns out to be well controlled, extraction can proceed today, either by the candidate or a specialist, with the usual risks of oroantral communication or root displacement into the sinus discussed given the tooth's position; if his HbA1c comes back high, or the result isn't available today, the safer course is to relieve the acute infection with pulp extirpation rather than a full extraction, since the source of infection cannot be left untreated regardless; and if it emerges he has not been taking his diabetes medication at all, referral to hospital for medical stabilisation first would be the appropriate course. On antibiotics, the candidate would continue the amoxicillin-clavulanate his GP already started for a further two days to complete an adequate course, or, if he had been given amoxicillin alone, would add metronidazole provided he is not drinking alcohol. They would arrange imaging — an OPG, with a contrast CT of the neck if deep space involvement is suspected — and would explain that definitive surgical management, comprising extraction of the causative tooth and incision and drainage if fluctuance is present, would be carried out once he is medically stabilised and his blood glucose is optimised, ideally in a hospital operating theatre with appropriate anaesthetic cover if his risk profile warrants it. Finally, they would obtain informed consent, explaining the diagnosis, the proposed treatment, and the risks involved, including airway compromise, spread to the deep neck spaces, sepsis, and the anaesthetic risks specifically related to his diabetes.

Management of Cellulitis in an Asthmatic Patient PQ-31

Periodontics · 11 minutes · Cluster 3

The station. Mr Thomas Edison 21 years old is attending your surgery today, complaining of pain and swelling associated with one of his teeth on the top left side. Upon examination, you find that he has facial swelling, swollen lymph nodes and has a fever. Xray shows deep caries related to that tooth, with complex root morphology, roots of the root look in close approximation to the maxillary sinus. Patient has an appointment with the oral surgeon after 3 days for the removal of this tooth. Medical history given, allergy to penicillin and has Asthma. Manage patients' concerns and give an appropriate treatment plan.

An excellent candidate would begin by warmly greeting Mr Edison and establishing rapport before taking a focused history. They would clarify the nature of his penicillin allergy specifically — establishing that he

developed an isolated rash as a child, with no history of anaphylaxis, urticaria, angioedema, or bronchospasm — and would recognise this as a low-risk reaction rather than a true contraindication to all penicillins. They would assess his asthma control, confirming he uses Seretide twice daily and Ventolin as required, and would note his increased reliever use over the previous day as a sign of currently suboptimal control. A thorough clinical examination would follow: recording his temperature (38.2°C), assessing the extent of facial swelling, checking for trismus, testing his ability to swallow, and evaluating for any signs of airway compromise. The candidate would recognise this as spreading odontogenic cellulitis with systemic involvement, with the fever indicating the swelling is not yet stable. They would explain clearly to Mr Edison that the infection has spread beyond the tooth itself, and that while oral antibiotics can help contain it, the source of the infection ultimately needs to be removed. They would explain that removing the tooth today, before his scheduled specialist appointment in three days, carries its own complications given the current infection — the area may be difficult to fully numb, his mouth opening is reduced, the roots are complex, and there is a risk of sinus involvement given their proximity to the maxillary sinus floor. The candidate would present the two management pathways with their respective advantages and disadvantages: if the swelling remains stable, the preferred course is antibiotics now and extraction at the specialist appointment in three days as planned; if the swelling becomes unstable, hospital admission is preferred, where the medical team would assess his airway, take a culture, and start intravenous antibiotics, while the dental team would extract the tooth and place a drain. Throughout, the candidate would acknowledge Mr Edison's concerns about missing work with genuine empathy while clearly communicating the clinical urgency of the situation. They would provide health promotion advice and clear safety-netting instructions for the period while he waits for his specialist appointment, including the specific signs that should prompt him to seek urgent care sooner.

Communication & ethics

Dietary Advice for an Athletic Patient PQ-18

Communication & ethics · 11 minutes · Cluster 3

The station. Mr Sam Smith is a 28-year-old gym trainer. You have already done the examination in the last appointment and noticed several interproximal caries and erosion. You also noticed that his saliva is thick and stringy. You gave him oral hygiene instructions, advised to get some restorations and to fill the diet chart. He would like to know how to modify his diet. He is very health conscious, following good diet regime. Diet chart given: He is drinking lots of sports drink, coffee and tea with sugar, cereal bars, fizzy drinks fruits. He is fit and healthy. Answer patients concern and inform what are the changes he requires to implement. No need to give discuss treatment plan for caries.

An excellent candidate would begin by warmly welcoming Mr Smith and acknowledging his health-conscious lifestyle before reviewing the completed diet chart in a non-judgmental way. They would take a structured dietary history, clarifying the frequency, timing, and type of foods and drinks consumed — noting the sports drinks, flavoured fizzy drinks, coffee and tea with sugar, cereal bars, and fruit. confirming which medications he uses. The candidate would explain, in plain language, that sports drinks and fizzy drinks are both high in sugar and highly acidic, and that sipping them throughout the day — rather than consuming them in one sitting — dramatically increases the time teeth are under acid attack, contributing to both decay and enamel erosion. They would advise switching to plain water as the primary hydration source and limiting sports drinks to during intense exercise only, followed by a plain water rinse. The candidate would highlight that cereal bars and dried fruit, despite their healthy image, are sticky and high in fermentable sugars, and suggest alternatives such as fresh fruit, cheese, unsalted nuts, or yoghurt. They would recommend reducing sugar in tea and coffee, or switching to sugar-free options. The candidate would recommend using fluoride 5000ppm toothpaste twice daily, waiting 30 minutes after eating before brushing, and consider recommending a remineralising agent such as Tooth Mousse given the thick, stringy saliva suggesting reduced salivary flow. They would check Mr Smith's understanding using the teach-back method, acknowledge his motivation, and arrange a review appointment. and recommend meeting a sport nutritionist and reading the labels

recommend review 3monthly and flouride varnish

Antibiotic Prophylaxis in Patient with Joint Replacement

PQ-36

Communication & ethics · 11 minutes · Cluster 3

The station. Mark Danial is a new patient in your surgery. He is seeing you today, complaining about his upper canine 23 tooth, which is extremely mobile due to bone loss. PA of 23,24,25 was given with 23 severe bone loss (there is bone loss around 24 and 25 as well). 23 roots are close to the anterior wall of the sinus. You decided to extract 23. Patient had joint replacement 6 years ago. He is Allergic to penicillin. He wants antibiotic prophylaxis as his previous dentist was giving him before dental treatment. He is also having asthma and hypertension. Explain the management plan to the pt.

An excellent candidate would begin by warmly greeting Mr Danial understanding his concern and explaining him the findings first discuss the xray and explore his expectations

taking history about his joint surgery They would confirm the type of joint replaced, clarify that it was performed six years ago, ask whether there have been any complications, infections, or revisions since surgery, and screen for immunocompromising conditions including diabetes, steroid use, or immunosuppressive medications. any prologed hospital stay . They would note his penicillin allergy and his diagnoses of asthma and hypertension.

acknowledging his concern about antibiotic prophylaxis before proceeding to take a focused medical history. explore antibiotic allergy type if it was confirmed. make them understand that unnecessary use of antibiotic can lead to formation of superbugs which stop responding to the medicine and due to his allergy his options are anyways limited. Superbugs can be passed on to family members as well

Management i would want to be guided by your specialist if we can get in touch with him they can provide a verbal consent im happy to give you antibiotic if we cannot get in touch with them today i can splint and temporise this tooth for you (variation if the patient qualifies for Antibiotic say ill give it this time but ill pput it in your record that next time we need a lettter from specialist)

The candidate would then explain the planned procedure — extraction of 23, noting the proximity to the maxillary sinus and the need for careful technique — and address the antibiotic prophylaxis question directly and empathetically. They would explain that current Australian guidelines, consistent with the ADA and AAOS, do not recommend routine antibiotic prophylaxis for patients with uncomplicated joint replacements in immunocompetent patients, particularly when the replacement is more than two years old with no history of infection or revision. They would validate Mark's previous experience while gently clarifying that the evidence base has evolved.

The candidate would acknowledge that prophylaxis may be appropriate in specific circumstances - in immunocompromised patients, or those with a history of prosthetic joint infection or those whom the surgeon has given a letter mentioning need of antibiotic prophylaxis— and offer to liaise with Mark's orthopaedic surgeon if he remained concerned. They would confirm his understanding, invite questions, and document the discussion. Throughout, they would maintain a calm, empathetic, and evidence-based approach.

health promotion GP referral to confirm allergy daily bacterimia if you dont maintain hygiene carries more risk of infection than the procedures we do 3monthly review

Drug-Seeking Behaviour — Opioid Request for Dental Pain

PQ-58

Communication & ethics · 11 minutes · Cluster 1

The station. *Female patient. Tooth known to need RCT — previous dentist diagnosed irreversible pulpitis and recommended root canal treatment. Scenario: Patient presents demanding strong prescription painkillers (specifically fentanyl, oxycodone, or tramadol) for toothache. Claims her regular dentist is away and she cannot contact him. States normal painkillers are not effective. She also has a physiotherapist-prescribed pain relief patch for chronic back pain. Has an appointment with her regular dentist next week. Insisting she needs strong medication urgently. Key Complexity: Drug-seeking behaviour. Task: Manage the patient's request. Address her concerns without prescribing opioids. Provide safe...*

This station tests the candidate's ability to recognise drug-seeking behaviour, maintain professional boundaries, and manage acute dental pain appropriately without prescribing opioids. On presentation, several red flags are immediately apparent: the patient is requesting specific opioids by name (fentanyl, oxycodone, tramadol), claims her regular dentist is unavailable, states that standard analgesics are ineffective, and already has an opioid-adjacent medication (fentanyl patch) prescribed for a separate condition. These cues collectively raise concern for drug-seeking behaviour or opioid misuse. The candidate should acknowledge the patient's pain empathetically and non-judgementally, validating that irreversible pulpitis is genuinely painful. However, the candidate must clearly and calmly decline to prescribe opioids, explaining that opioids are not first-line dental analgesics, carry significant risks including dependence and interaction with her existing patch, and are not indicated when definitive dental treatment is available. Appropriate management includes prescribing or recommending a combination of ibuprofen 400 mg and paracetamol 500–1000 mg alternating every 3–4 hours, which has evidence-based efficacy superior to opioids for dental pain. If infection is suspected, antibiotics such as amoxicillin should be considered. The candidate should offer an urgent dental referral or same-day emergency dental appointment rather than pharmacological substitution. Safety-netting includes advising the patient to present to an emergency department if she develops systemic signs of infection, trismus, or swelling. The refusal to prescribe opioids, the clinical rationale, and the alternative management plan must be clearly documented in the patient record.